EN.553.733 Nonparametric Bayesian Statistics

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Contents

Chapter 1

Introduction

The Bayesian point of view with y data and fixed θ paramaters:

$$\mathbb{P}(\theta|y) \propto \mathbb{P}(y|\theta)\mathbb{P}(\theta)$$

Nonparametric Bayes (BNP) extends this by allowing unbounded/growing/infinite parameters

1.1 Exchangeability and De Finetti's Theorem

A theoretical motivation for nonparametric Bayesian methods comes from *De Finetti's Theorem*.

Exchangeability

A sequence $(X_n)_{n\geq 1}$ is *infinitely exchangeable* if for every N and every permutation σ of $\{1,\ldots,N\}$,

$$p(X_1,\ldots,X_N) = p\Big(X_{\sigma(1)},\ldots,X_{\sigma(N)}\Big).$$

This means the joint distribution depends only on the multiset of observations, not their order.

De Finetti Representation

Theorem 1.1.1 (De Finetti (informal))**.** *A sequence $X_1, X_2, \ldots$ is infinitely exchangeable if and only if, for all N, there exists a probability measure P such that*

$$p(X_1,\ldots,X_N) = \int_\Theta \prod_{n=1}^{N} p(X_n \mid \theta)\, P(d\theta).$$

Conditionally on θ, the observations are i.i.d., i.e.,

$$X_n \mid \theta \overset{i.i.d.}{\sim} p(\cdot \mid \theta).$$

Thus exchangeability implies a mixture-of-i.i.d. representation.

Implications

This representation motivates:

- likelihood models $p(x \mid \theta)$,
- prior distributions $P(d\theta)$,
- *nonparametric Bayesian* priors, where P is infinite-dimensional.

1.2 The Bayesian Toolkit

1.2.1 Likelihood

At the core of Bayesian inference is the *likelihood*. Given data $D = x = (x_1, \ldots, x_n)$,

$$L(\theta \mid D) = f(x \mid \theta).$$

If $x_1, \ldots, x_n$ are conditionally independent given θ, then

$$f(x \mid \theta) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

1.2.2 Prior Distribution

Uncertainty about θ is encoded through a *prior* $\pi(\theta)$.

1.2.3 Joint, Marginal, and Posterior

Joint distribution

$$\psi(\theta, x) = f(x \mid \theta)\pi(\theta).$$

Marginal distribution

$$m(x) = \int f(x \mid \theta)\pi(\theta)\, d\theta.$$

This quantity is also called the *evidence* or *model likelihood*.

Posterior distribution

$$\pi(\theta \mid x) = \frac{f(x \mid \theta)\pi(\theta)}{\int f(x \mid \theta)\pi(\theta)\, d\theta} = \frac{f(x \mid \theta)\pi(\theta)}{m(x)}.$$

1.2.4 Predictive Distribution

If $y \sim g(y \mid \theta, x)$, then the predictive distribution is

$$g(y \mid x) = \int g(y \mid \theta, x)\, \pi(\theta \mid x)\, d\theta.$$

This integrates parameter uncertainty into predictions.

Remark While inference focuses on $\pi(\theta \mid x)$, prediction via $g(y \mid x)$ is often the primary goal.

1.3 Conjugate Priors

1.3.1 Definition

Theorem 1.3.1 (Conjugate Prior)**.** *A class $\mathcal{P}$ of prior distributions is conjugate for a likelihood $p(x \mid \theta)$ if*

$$\pi(\theta) \in \mathcal{P} \;\Rightarrow\; \pi(\theta \mid x) \in \mathcal{P}.$$

Examples

- If $x \mid \theta \sim \mathrm{Bin}(n, \theta)$ and $\theta \sim \mathrm{Beta}(a, b)$, then

$$\theta \mid x \sim \mathrm{Beta}(a + x,\ b + n - x).$$

- If $x_1, \ldots, x_n \overset{i.i.d.}{\sim} \mathrm{Poisson}(\lambda)$ and $\lambda \sim \mathrm{Gamma}(a, b)$, then

$$\lambda \mid x \sim \mathrm{Gamma}\Big(a + \textstyle\sum_{i=1}^{n} x_i,\ b + n\Big).$$

1.3.2 Properties

Advantage *Closed-form* posterior updates.

Disadvantage The conjugate family may not reflect prior beliefs.

1.4 Credible Intervals

Theorem 1.4.1 (Credible Interval)**.** *An interval* (a, b) *is a* $(1 - \alpha)100\%$ *credible interval for* θ *if*

$$\Pr(a < \theta < b \mid x) = 1 - \alpha.$$

Quantile Construction

Choose a, b such that

$$\Pr(\theta < a \mid x) = \frac{\alpha}{2}, \qquad \Pr(\theta > b \mid x) = \frac{\alpha}{2}.$$

Remark

Credible intervals quantify posterior uncertainty, unlike frequentist confidence intervals which are defined via repeated sampling.

1.5 MAP Estimator

The *maximum a posteriori (MAP)* estimator is defined as

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} f(x \mid \theta)\pi(\theta).$$

Equivalently,

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \left[\log f(x \mid \theta) + \log \pi(\theta)\right].$$

Motivation

- Interpretable as a *penalized likelihood* estimator.
- Useful when optimization is easier than integration.

1.6 Testing Hypotheses

Hypothesis testing on θ is formulated as

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \notin \Theta_0.$$

1.7 Bayes Factor

Bayesian testing procedures can be based on the posterior probability

$$P^\pi(\theta \in \Theta_0 \mid x),$$

or alternatively on the *Bayes factor*.

Definition

The Bayes factor comparing Θ_1 to Θ_0 is

$$B_{10}^\pi = \frac{P^\pi(\theta \in \Theta_1 \mid x)/P^\pi(\theta \in \Theta_0 \mid x)}{P^\pi(\theta \in \Theta_1)/P^\pi(\theta \in \Theta_0)}.$$

Model Comparison Formulation

For corresponding models $\mathcal{M}_1$ versus $\mathcal{M}_0$,

$$B_{10}^\pi = \frac{P^\pi(\mathcal{M}_1 \mid x)}{P^\pi(\mathcal{M}_0 \mid x)} \bigg/ \frac{P^\pi(\mathcal{M}_1)}{P^\pi(\mathcal{M}_0)}.$$

If the prior is written as

$$\pi(\theta) = \Pr(\theta \in \Theta_1)\, \pi_1(\theta) + \Pr(\theta \in \Theta_0)\, \pi_0(\theta),$$

then

$$B_{10}^\pi = \frac{\int f(x \mid \theta_1)\pi_1(\theta_1)\, d\theta_1}{\int f(x \mid \theta_0)\pi_0(\theta_0)\, d\theta_0} = \frac{m_1(x)}{m_0(x)}.$$

Thus the Bayes factor is *akin to a likelihood ratio.*

1.8 Jeffreys' Scale

Interpretation of the Bayes factor is often based on $\log_{10}(B_{10}^{\pi})$:

- if $\log_{10}(B_{10}^{\pi}) \in [0, 0.5]$, evidence against H_0 is poor,
- if $\log_{10}(B_{10}^{\pi}) \in [0.5, 1]$, it is substantial,
- if $\log_{10}(B_{10}^{\pi}) \in [1, 2]$, it is strong,
- if $\log_{10}(B_{10}^{\pi}) > 2$, it is decisive.

1.9 Markov Chain Monte Carlo (MCMC)

Statistical inference is based on the posterior distribution

$$\pi(\theta \mid x) = \frac{\pi(\theta) f(x \mid \theta)}{\int \pi(\theta) f(x \mid \theta)\, d\theta}.$$

Motivation

- In conjugate cases (e.g. normal-normal, binomial-beta), integrals can be computed analytically.
- In general, closed-form expressions for $\pi(\theta \mid x)$ are not available.
- We still need to compute:
  - point estimates,
  - interval estimates,
  - hypothesis tests.

This motivates simulation-based methods.

The MCMC Principle

Definition

An *MCMC method* for simulating from a distribution π is any method that constructs an ergodic Markov chain (X_n) whose stationary distribution is π.

Given an initial value $x^{(0)}$, the chain is generated via a transition kernel such that

$$X_n \xrightarrow{d} X \sim \pi.$$

Construction

To build such a chain, common methods include:

- Metropolis–Hastings,

- Gibbs sampling.

There are also diagnostic methods to assess whether the chain has reached its stationary distribution.

Metropolis–Hastings Algorithm

Algorithm

Given current state $x^{(t)}$:

- Generate proposal $Y_t \sim q(y \mid x^{(t)})$ (*proposal distribution*).
- Set
$$X^{(t+1)} = \begin{cases} Y_t, & \text{with probability } \rho(x^{(t)}, Y_t), \\ x^{(t)}, & \text{with probability } 1 - \rho(x^{(t)}, Y_t), \end{cases}$$
where
$$\rho(x, y) = \min\left\{ \frac{\pi(y)\, q(x \mid y)}{\pi(x)\, q(y \mid x)},\ 1 \right\}.$$

Remarks on Metropolis–Hastings

- The quantity $\rho(x, y)$ is the *acceptance ratio*.
- The resulting chain is Markov since $X^{(t+1)}$ depends only on $X^{(t)}$.
- If $q(x \mid y) = q(y \mid x)$, the proposal is symmetric and the method reduces to the *Metropolis algorithm*.
- In this case, the acceptance probability simplifies to
$$\frac{\pi(y_t)}{\pi(x^{(t)})}.$$

1.10 Additional Remarks

Acceptance Mechanism

If
$$\frac{\pi(y_t)}{q(y_t \mid x^{(t)})} > \frac{\pi(x_t)}{q(x^{(t)} \mid y_t)},$$
the proposal is always accepted. Otherwise, it is accepted with probability ρ.

Invariance

The algorithm depends only on ratios:
$$\frac{\pi(y_t)}{\pi(x^{(t)})}, \qquad \frac{q(x^{(t)} \mid y_t)}{q(y_t \mid x^{(t)})}.$$

Thus, normalizing constants cancel out, provided they do not depend on the conditioning variables.

Choice of Proposal

The choice of proposal distribution q strongly affects performance (efficiency and convergence).

A good proposal should:

- be easy to sample from,
- allow easy computation of the acceptance ratio,
- move a reasonable distance in parameter space,
- not be rejected too frequently.

1.11 Gibbs Sampling

1.11.1 Overview

The Gibbs sampler is one of the most widely used computational methods for Bayesian inference. It was first used in statistical physics and later popularized in statistics.

The method is particularly useful for high-dimensional target distributions.

1.11.2 Key Idea

The Gibbs sampler works by *sequentially sampling from univariate conditional distributions* rather than attempting to sample directly from a multivariate distribution.

1.11.3 Two-Stage Gibbs Sampler

Consider a bivariate posterior distribution

$$\pi(\theta, \lambda \mid x).$$

Algorithm

- **Initialization:** Start with an arbitrary value $\lambda^{(0)}$.
- **Iteration t:**
  - Sample
  $$\theta^{(t)} \sim \pi_1(\theta \mid x, \lambda^{(t-1)}),$$
  - Sample
  $$\lambda^{(t)} \sim \pi_2(\lambda \mid x, \theta^{(t)}).$$

The joint distribution $\pi(\theta, \lambda \mid x)$ is a stationary distribution for this Markov chain.

1.11.4 Multi-Stage Gibbs Sampler

General Form

For a joint distribution $\pi(\theta \mid x)$ with full conditionals $\pi_1, \ldots, \pi_p$, define the iterative scheme:

Given $(\theta_1^{(t)}, \ldots, \theta_p^{(t)})$,

$$\begin{aligned}
\theta_1^{(t+1)} &\sim \pi_1(\theta_1 \mid x, \theta_2^{(t)}, \ldots, \theta_p^{(t)}), \\
\theta_2^{(t+1)} &\sim \pi_2(\theta_2 \mid x, \theta_1^{(t+1)}, \theta_3^{(t)}, \ldots, \theta_p^{(t)}), \\
&\vdots \\
\theta_p^{(t+1)} &\sim \pi_p(\theta_p \mid x, \theta_1^{(t+1)}, \ldots, \theta_{p-1}^{(t+1)}).
\end{aligned}$$

Then

$$\theta^{(t)} \;\to\; \theta \sim \pi(\theta \mid x).$$

1.11.5 Remarks on Gibbs Sampling

- The order of updates is not essential and can be changed without affecting validity.
- The update order can also be randomized (*random scan Gibbs*).
- It is not necessary to update every component at each iteration, provided each component is updated sufficiently often.
- The overall process defines a Markov chain, but the individual component sequences $\theta_i^{(t)}$ are generally not Markov.

1.11.6 Convergence Diagnostics

An MCMC algorithm is said to have converged at iteration T if for all $t > T$, the samples can be regarded as drawn from the stationary distribution.

- Lack of convergence leads to biased and unreliable estimates.
- Diagnostics are used to detect signs of non-convergence.
- Passing a diagnostic does *not* guarantee stationarity.

Common Diagnostics

- Geweke
- Gelman and Rubin
- Heidelberger and Welch
- Raftery and Lewis

See the *R package CODA* for implementation.

Quick Checks of Convergence

- A *trace plot* of $\theta^{(t)}$ against t provides a basic visual diagnostic.
- Chains may appear to converge if trapped near a local mode.
- Running multiple chains from different starting values helps assess convergence.

1.12 Autocorrelation

Mixing

The speed at which the chain explores the parameter space is called *mixing*.

High autocorrelation implies slow mixing and slow convergence.

Autocorrelation Function

For a sequence $\{\theta^{(1)}, \ldots, \theta^{(T)}\}$, the lag-$t$ autocorrelation function is

$$\mathrm{acf}_t(\theta) = \frac{\frac{1}{T-t}\sum_{j=1}^{T-t}(\theta^{(j)} - \bar{\theta})(\theta^{(j+t)} - \bar{\theta})}{\frac{1}{T-t}\sum_{j=1}^{T}(\theta^{(j)} - \bar{\theta})^2}.$$

Remarks

- Gibbs sampling offers limited control over correlation structure.
- In Metropolis-type algorithms, correlation can be adjusted through the proposal distribution.
- Adaptive variants modify the proposal to improve mixing.

Practical Implications

- High autocorrelation indicates slow convergence.
- Thinning may be used to reduce correlation between retained samples.

1.13 Reading: Mixture Models and Bayesian Computation

Motivation for Mixtures

Mixture models provide a flexible framework for modelling complex data distributions:

$$f(x) = \sum_{i=1}^{k} p_i f(x \mid \theta_i), \quad \sum_{i=1}^{k} p_i = 1.$$

They serve two key roles:

- *Latent structure modelling*: data arise from multiple subpopulations (clustering interpretation)
- *Approximation*: mixtures act as flexible approximations to unknown densities

Thus, mixtures bridge parametric and nonparametric modeling.

Latent Variable Representation

Introduce allocation variables

$$Z_i \sim \text{Multinomial}(1; p_1, \ldots, p_k), \quad X_i \mid Z_i = j \sim f(x \mid \theta_j).$$

This transforms the model into a *complete-data formulation*, which is critical for computation.

Computational Challenge

The likelihood for n observations is

$$L(\theta, p \mid x) = \prod_{i=1}^{n} \sum_{j=1}^{k} p_j f(x_i \mid \theta_j).$$

Expanding this produces k^n terms, making direct inference intractable.
Implication: Closed-form Bayesian inference is generally impossible.

Combinatorial Explosion

Posterior inference can be written as

$$\pi(\theta, p \mid x) = \sum_{z} \omega(z)\, \pi(\theta, p \mid x, z),$$

where z are allocation vectors.

- Number of allocations: k^n
- Only a *tiny subset* have non-negligible weight

This motivates stochastic approximation methods (MCMC).

EM vs Bayesian Computation

The EM algorithm alternates:

- E-step: compute expected allocations
- M-step: maximize expected complete likelihood

While efficient for MLE, EM:

- converges only to *local modes*

- is sensitive to initialization

Key contrast:

- EM: optimization
- MCMC: full posterior exploration

Mixtures as Ill-Posed Problems

Mixture inference is inherently unstable:

- Some components may receive *no data*
- Likelihood can become *unbounded*
- Small data changes $\Rightarrow$ large parameter changes

This is a classic *inverse problem*.

Label Switching and Identifiability

The likelihood is invariant under permutation:

$$(\theta_1, \ldots, \theta_k) \sim (\theta_{\sigma(1)}, \ldots, \theta_{\sigma(k)}).$$

Consequences:

- Posterior has $\mathcal{O}(k!)$ symmetric modes
- Marginal posteriors for components are identical
- Standard estimators (means) are meaningless

This is known as the *label switching problem*.

Prior Design in Mixtures

Improper priors are problematic:

$$\pi(\theta) = \prod_i \pi_i(\theta_i) \quad \Rightarrow \quad \text{posterior may be improper.}$$

Reason:

- Some components may have no observations
- Their posterior remains equal to the prior

Solution:

- Use proper priors
- Introduce dependence between components

Mixtures and Nonparametric Bayes

Mixtures naturally connect to nonparametric Bayesian methods:

- Kernel density estimation:
$$\hat{f}(x) = \frac{1}{n}\sum_{i=1}^{n} K(x - x_i)$$
is a mixture model
- Dirichlet process:
$$F \sim \mathrm{DP}(F_0, \alpha)$$
induces an *infinite mixture representation*
- Bernstein polynomials:
$$f(x) = \sum_k p_k \mathrm{Beta}(\alpha_k, \beta_k)$$
approximate arbitrary densities

Thus, Bayesian nonparametrics can be viewed as *mixtures with infinitely many components*.

Key Takeaways for MCMC

- Mixture models are computationally intractable in closed form
- Latent variables enable tractable conditional structure
- MCMC (Gibbs, MH) exploits this structure
- Posterior geometry is multimodal and symmetric
- Careful diagnostics and interpretation are required

Chapter 2

Dirichlet Process

Bayesian Models Revisited

A Bayesian statistical model consists of:

- a likelihood $f(y \mid \theta)$,
- a prior distribution $\pi(\theta)$.

Uncertainty about parameters θ is modeled through the prior distribution. Inference is based on the posterior distribution

$$\pi(\theta \mid y) = \frac{f(y \mid \theta)\pi(\theta)}{\int f(y \mid \theta)\pi(\theta)\, d\theta}.$$

2.1 Nonparametric Bayesian Priors

Modeling Idea

A *nonparametric Bayesian model* places a distribution on distributions.

$$\theta \mid G \sim G$$

where G itself is random:

$$G \sim P(G).$$

Thus, uncertainty is modeled at two levels:

- uncertainty in θ,
- uncertainty in the distribution G.

Inference may target both θ and G.

2.2 Finite Mixture Model

Generative Model (Two Components)

Consider a Gaussian mixture model with $K = 2$:

- Latent assignments:
$$z_n \overset{iid}{\sim} \text{Categorical}(\rho_1, \rho_2),$$
- Observations:
$$x_n \mid z_n \sim \mathcal{N}(\mu_{z_n}, \Sigma),$$
- Component means:
$$\mu_k \overset{iid}{\sim} \mathcal{N}(\mu_0, \Sigma_0),$$
- Mixing weights:
$$\rho_1 \sim \text{Beta}(a_1, a_2), \quad \rho_2 = 1 - \rho_1.$$

Inference Goal

- assign data points to clusters,
- estimate cluster parameters.

2.3 Beta Distribution Review

The Beta distribution is

$$\text{Beta}(\rho_1 \mid a_1, a_2) = \frac{\Gamma(a_1 + a_2)}{\Gamma(a_1)\Gamma(a_2)} \rho_1^{a_1-1}(1-\rho_1)^{a_2-1}.$$

Gamma Function

- For integer m: $\Gamma(m + 1) = m!$
- For $x > 0$: $\Gamma(x + 1) = x\Gamma(x)$

Behavior

- $a_1 = a_2 \to 0$: mass near boundaries
- $a_1 = a_2 \to \infty$: concentration around 1/2
- $a_1 > a_2$: skew toward 1

Conjugacy

If

$$\rho_1 \sim \text{Beta}(a_1, a_2), \quad z \sim \text{Categorical}(\rho_1, \rho_2),$$

then

$$\rho_1 \mid z \sim \text{Beta}(a_1 + \mathbf{1}\{z = 1\},\ a_2 + \mathbf{1}\{z = 2\}).$$

2.4 Dirichlet Distribution

Definition

For $\rho_{1:K}$,

$$\text{Dirichlet}(\rho_{1:K} \mid a_{1:K}) = \frac{\Gamma\left(\sum_{k=1}^{K} a_k\right)}{\prod_{k=1}^{K} \Gamma(a_k)} \prod_{k=1}^{K} \rho_k^{a_k - 1},$$

with

$$\rho_k > 0, \quad \sum_{k=1}^{K} \rho_k = 1.$$

Construction via Gamma

Let

$$Z_j \sim \text{Gamma}(a_j, 1), \quad j = 1, \dots, K,$$

and define

$$Y_j = \frac{Z_j}{\sum_{l=1}^{K} Z_l}.$$

Then

$$(Y_1, \dots, Y_K) \sim \text{Dirichlet}(a_1, \dots, a_K).$$

Special Case

For $K = 2$:

$$\text{Dirichlet}(a_1, a_2) = \text{Beta}(a_1, a_2).$$

Moments

$$E(Y_j) = \frac{a_j}{\sum_{l=1}^{K} a_l}, \quad E(Y_j^2) = \frac{a_j(a_j + 1)}{(\sum_l a_l)(1 + \sum_l a_l)}.$$

For $i \neq j$:

$$E(Y_i Y_j) = \frac{a_i a_j}{(\sum_l a_l)(1 + \sum_l a_l)}.$$

Marginals

$$Y_j \sim \text{Beta}\left(a_j,\ \sum_{i\neq j} a_i\right).$$

Conjugacy

If

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}), \quad z \sim \text{Categorical}(\rho_{1:K}),$$

then

$$\rho_{1:K} \mid z \sim \text{Dirichlet}(a'_1, \ldots, a'_K), \quad a'_k = a_k + \mathbf{1}\{z = k\}.$$

2.5 Finite Mixture Model (General K)

- Mixing weights:
$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}),$$
- Component means:
$$\mu_k \sim \mathcal{N}(\mu_0, \Sigma_0),$$
- Assignments:
$$z_n \sim \text{Categorical}(\rho_{1:K}),$$
- Observations:
$$x_n \mid z_n \sim \mathcal{N}(\mu_{z_n}, \Sigma).$$

2.6 When $K > N$

Interpretation

- K: number of possible latent groups (components)
- clusters: number of components actually represented in the data

Key Observations

- Number of clusters is random and typically less than K
- Number of clusters grows with sample size N

Motivation

Choosing a finite K is difficult:

- K is unknown in practice
- inference depends heavily on K
- problematic for streaming or growing datasets

2.7 Choosing $K = \infty$

Motivation

In mixture models, choosing a finite K is difficult:

- K is unknown a priori,
- inference depends strongly on K,
- problematic for streaming or growing datasets.

This motivates considering the limit

$$K = \infty.$$

Infinite Mixing Weights

We seek a sequence of strictly positive weights

$$\rho = (\rho_1, \rho_2, \ldots) \quad \text{such that} \quad \sum_{k=1}^{\infty} \rho_k = 1.$$

Dirichlet Decomposition

From the finite Dirichlet:

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}),$$

we have the decomposition:

$$\rho_1 \sim \text{Beta}\Big(a_1,\ \sum_{k=2}^{K} a_k\Big),$$

and conditionally,

$$\frac{(\rho_2, \ldots, \rho_K)}{1 - \rho_1} \sim \text{Dirichlet}(a_2, \ldots, a_K).$$

This suggests a recursive construction.

2.8 Stick-Breaking Construction

2.8.1 Finite Construction

Define:

$$V_1 \sim \mathrm{Beta}(a_1, a_2 + a_3 + a_4), \quad \rho_1 = V_1,$$
$$V_2 \sim \mathrm{Beta}(a_2, a_3 + a_4), \quad \rho_2 = (1 - V_1)V_2,$$
$$V_3 \sim \mathrm{Beta}(a_3, a_4), \quad \rho_3 = (1 - V_1)(1 - V_2)V_3,$$

and so on, with

$$\rho_K = 1 - \sum_{k=1}^{K-1} \rho_k.$$

This is the *stick-breaking* representation.

2.8.2 Infinite Stick-Breaking

Taking $K \to \infty$, define

$$V_k \sim \mathrm{Beta}(a_k, b_k),$$

and construct weights:

$$\rho_k = \left(\prod_{j=1}^{k-1} (1 - V_j) \right) V_k.$$

2.8.3 GEM Distribution

A special case of stick-breaking is:

$$a_k = 1, \quad b_k = \alpha > 0.$$

Then

$$\rho = (\rho_1, \rho_2, \ldots) \sim \mathrm{GEM}(\alpha),$$

with

$$V_k \sim \mathrm{Beta}(1, \alpha), \quad \rho_k = \left(\prod_{j=1}^{k-1} (1 - V_j) \right) V_k.$$

This defines a random probability distribution over a countably infinite set.

2.9 Interpretation

Hierarchy of Distributions

- Beta: random distribution over $\{1, 2\}$
- Dirichlet: random distribution over $\{1, \ldots, K\}$
- GEM / stick-breaking: random distribution over $\{1, 2, \ldots\}$

Key Properties

- Infinitely many components
- Only finitely many used for finite data
- Number of active components grows with N

Conceptual Insight

- Parameters: infinite mixture components
- Effective parameters: number of occupied clusters

2.10 Readings: Foundations of the Dirichlet Process

Dirichlet Process as a Distribution over Distributions

A Dirichlet process defines a *random probability measure* P on a measurable space $(\mathcal{X}, \mathcal{B})$.

- P itself is the parameter of interest
- P takes values in the space of all probability measures

The key defining property is:
For any measurable partition $(B_1, \ldots, B_k)$,

$$\Big(P(B_1), \ldots, P(B_k)\Big) \sim \text{Dirichlet}\Big(a(B_1), \ldots, a(B_k)\Big),$$

where a is a finite base measure.

Interpretation of the Base Measure

Let

$$a(\mathcal{X}) = \alpha, \quad \beta(\cdot) = \frac{a(\cdot)}{\alpha}.$$

Then:

- β is the *base distribution* (mean measure)
- α controls concentration

Key intuition:

$$E[P(B)] = \beta(B)$$

Thus, the Dirichlet process is centered around β.

Discrete Nature of the Dirichlet Process

A fundamental property:

- With probability 1, P is *discrete*

  That is,

  $$P = \sum_{n=1}^{\infty} p_n \delta_{Y_n}$$

  even if β is continuous.
  This is crucial:

- induces clustering
- explains repeated parameter values

Posterior Conjugacy

If

$$P \sim \mathrm{DP}(a), \quad X \mid P \sim P,$$

then the posterior is:

$$P \mid X \sim \mathrm{DP}(a + \delta_X).$$

For data $(X_1, \ldots, X_n)$:

$$P \mid X_{1:n} \sim \mathrm{DP}\left(a + \sum_{i=1}^{n} \delta_{X_i}\right).$$

This is the infinite-dimensional analogue of Dirichlet conjugacy.

Constructive Representation (Stick-Breaking)

A constructive definition of P is:

$$P = \sum_{n=1}^{\infty} p_n \delta_{Y_n}$$

where:

- $Y_n \sim \beta$
- weights are given by stick-breaking:

  $$p_1 = V_1, \quad p_n = V_n \prod_{j=1}^{n-1} (1 - V_j)$$

- $$V_n \sim \mathrm{Beta}(1, \alpha)$$

This gives an explicit generative construction of the random measure.

Distributional Fixed-Point Property

The Dirichlet process satisfies the distributional equation:

$$P = \theta\delta_Y + (1-\theta)P',$$

where:

- $\theta \sim \text{Beta}(1, \alpha)$
- $Y \sim \beta$
- $P' \overset{d}{=} P$ and independent

This recursive structure underlies stick-breaking.

Connection to Bayesian Nonparametrics

The Dirichlet process enables:

- modeling unknown distributions directly
- infinite mixture models
- automatic complexity control

Instead of fixing K, we obtain:

$$K = \infty \quad \text{but effective clusters } < \infty.$$

Practical Insight

- Only finitely many atoms receive non-negligible mass
- Sampling requires only a finite truncation
- Posterior updates remain tractable

This resolves the key issue:

$$\text{Unknown number of clusters} \Rightarrow \text{handled probabilistically.}$$

Distributions Hierarchy Interpretation

We can view the progression:

- Beta: random distribution over $\{1, 2\}$
- Dirichlet: random distribution over $\{1, 2, \ldots, K\}$
- GEM / stick-breaking: random distribution over $\{1, 2, \ldots\}$
- Dirichlet process: random distribution over Φ

Formally, the Dirichlet process admits the representation:

$$\rho = (\rho_1, \rho_2, \dots) \sim \mathrm{GEM}(\alpha), \quad \phi_k \overset{i.i.d.}{\sim} G_0,$$

$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\phi_k}.$$

Thus:

- ρ_k controls cluster weights
- ϕ_k are atom locations
- G is a discrete random measure

2.10.1 Dirichlet Process Mixture Model (DPMM)

Gaussian mixture case

- Weights:
$$\rho \sim \mathrm{GEM}(\alpha)$$
- Cluster parameters:
$$\mu_k \overset{i.i.d.}{\sim} \mathcal{N}(\mu_0, \Sigma_0)$$
- Random measure:
$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\mu_k} \equiv DP(\alpha, \mathcal{N}(\mu_0, \Sigma_0))$$

Latent structure:

$$z_n \overset{i.i.d.}{\sim} \mathrm{Categorical}(\rho), \quad \mu_n^* = \mu_{z_n}, \quad \mu_n^* \overset{i.i.d.}{\sim} G$$

Observations:

$$x_n \mid \mu_n^* \sim \mathcal{N}(\mu_n^*, \Sigma)$$

General formulation

More generally:

$$\rho \sim \mathrm{GEM}(\alpha), \quad \phi_k \overset{i.i.d.}{\sim} G_0$$

$$G = \sum_{k=1}^{\infty} \rho_k \delta_{\phi_k} \equiv DP(\alpha, G_0)$$

$$z_n \sim \mathrm{Categorical}(\rho), \quad \theta_n = \phi_{z_n}, \quad \theta_n \overset{i.i.d.}{\sim} G$$

$$x_n \mid \theta_n \sim F(\theta_n)$$

Dirichlet Process Definition

Let $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$ be a measurable space.

Theorem 2.10.1. *(Dirichlet Process) A random probability measure F satisfies*

$$F \sim DP(\alpha, H)$$

if for any measurable partition $(B_1, \ldots, B_n)$,

$$(F(B_1), \ldots, F(B_n)) \sim \text{Dirichlet}(\alpha H(B_1), \ldots, \alpha H(B_n)).$$

Key implication:

$$p(F) \text{ is well-defined via Kolmogorov consistency.}$$

Properties of the Dirichlet Process

Moments

For any measurable A:

$$E[F(A)] = H(A), \quad \text{Var}(F(A)) = \frac{H(A)(1 - H(A))}{1 + \alpha}.$$

Self-similarity

For B such that $H(B) > 0$:

$$F_B \sim DP(\alpha, H_B),$$

where F_B is the restriction of F to B.

Conjugacy

If

$$y_i \overset{i.i.d.}{\sim} F, \quad F \sim DP(\alpha, H),$$

then

$$F \mid y \sim DP\left(\alpha + n,\ \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{y_i} + \frac{\alpha}{\alpha + n} H\right).$$

In particular:

$$E[F(A) \mid y] = \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{y_i}(A) + \frac{\alpha}{\alpha + n} H(A).$$

Richness

- Full support over distributions absolutely continuous w.r.t. H
- Can approximate arbitrary distributions

Stick-Breaking Theorem

Theorem 2.10.2. *(Sethuraman) Let*

$$\theta_k \overset{i.i.d.}{\sim} H, \quad V_k \overset{i.i.d.}{\sim} \mathrm{Beta}(1, \alpha),$$

and define

$$\pi_k = V_k \prod_{l=1}^{k-1} (1 - V_l).$$

Then

$$\sum_{k=1}^{\infty} \pi_k \delta_{\theta_k} \sim DP(\alpha, H).$$

Posterior Stick-Breaking Interpretation

Given

$$G \sim DP(\alpha, H), \quad \theta \sim G,$$

we obtain:

$$G \mid \theta \sim DP\left(\alpha + 1, \ \frac{\alpha H + \delta_\theta}{\alpha + 1}\right).$$

This implies a decomposition:

$$G = V\delta_\theta + (1 - V)G', \quad V \sim \mathrm{Beta}(1, \alpha),$$

where G' is the residual measure.

Recursive Property of the DP

For a partition $(\theta, A_1, \ldots, A_K)$:

$$(G(\theta), G(A_1), \ldots, G(A_K)) \sim \mathrm{Dirichlet}(1, \alpha H(A_1), \ldots, \alpha H(A_K)).$$

This implies:

$$G' \sim DP(\alpha, H),$$

showing the *self-similar recursive structure* of the Dirichlet process.

Key Insight

The Dirichlet process can be understood as:

- a distribution over discrete measures
- constructed via stick-breaking
- inducing clustering through shared atoms
- recursively decomposable via Beta splitting

This structure is what enables:

$$\text{infinite mixture models} \quad \Longleftrightarrow \quad \text{finite data-driven clustering.}$$

Stick-Breaking Expansion

We can recursively expand the Dirichlet process draw:

$$G \sim DP(\alpha, H)$$

$$G = V_1\delta_{\theta_1^*} + (1 - V_1)G_1$$

$$G = V_1\delta_{\theta_1^*} + (1 - V_1)\Big(V_2\delta_{\theta_2^*} + (1 - V_2)G_2\Big)$$

Continuing:

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*},$$

where

$$\pi_k = V_k \prod_{l=1}^{k-1}(1 - V_l), \quad V_k \sim \text{Beta}(1, \alpha), \quad \theta_k^* \sim H.$$

Interpretation

- Draws from a DP are *discrete* measures
- Mass is allocated via stick-breaking
- Atoms θ_k^* are drawn from the base distribution H

Marginalizing the Weights

Consider a simple case:

$$\rho_1 \sim \text{Beta}(a_1, a_2), \quad z_n \sim \text{Categorical}(\rho_1, \rho_2).$$

Integrating out ρ_1, we obtain:

$$p(z_n = 1 \mid z_{1:n-1}) = \frac{a_{1,n}}{a_{1,n} + a_{2,n}},$$

where

$$a_{1,n} = a_1 + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = 1\}, \quad a_{2,n} = a_2 + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = 2\}.$$

2.10.2 Pólya Urn Interpretation

This predictive rule corresponds to a Pólya urn:

- Choose a ball with probability proportional to its count
- Replace it and add another ball of the same color

Then:

$$\lim_{n\to\infty} \frac{\#\ \text{orange}}{\#\ \text{total}} \overset{d}{=} \text{Beta}(a_{\text{orange}}, a_{\text{green}}).$$

Multivariate Extension

For

$$\rho_{1:K} \sim \text{Dirichlet}(a_{1:K}), \quad z_n \sim \text{Categorical}(\rho_{1:K}),$$

we obtain:

$$p(z_n = k \mid z_{1:n-1}) = \frac{a_{k,n}}{\sum_{j=1}^{K} a_{j,n}},$$

where

$$a_{k,n} = a_k + \sum_{m=1}^{n-1} \mathbf{1}\{z_m = k\}.$$

This corresponds to a *multivariate Pólya urn*.

2.10.3 Blackwell–MacQueen Pólya Urn

We now consider the infinite limit.

Let:

$$G \sim DP(\alpha, H), \quad \theta_i \mid G \sim G.$$

Then:

First sample

$$\theta_1 \sim H, \quad G \mid \theta_1 \sim DP\left(\alpha + 1, \frac{\alpha H + \delta_{\theta_1}}{\alpha + 1}\right).$$

Second sample

$$\theta_2 \mid \theta_1 \sim \frac{\alpha H + \delta_{\theta_1}}{\alpha + 1}.$$

General step

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1}.$$

Urn Interpretation

- With probability proportional to α, draw a new value from H
- With probability proportional to existing counts, reuse a previous value

Equivalently:

- Draw a ball proportional to its mass
- If it is *new*, create a new color
- Otherwise, reinforce the existing color

Key Insight

This produces a sequence:

$$\theta_1, \theta_2, \ldots$$

with:

- clustering (shared values)
- increasing number of clusters
- probability of new cluster controlled by α

2.11 Chinese Restaurant Process (CRP)

Construction

The Chinese Restaurant Process provides an equivalent formulation of the Dirichlet process clustering mechanism.

- Each observation (customer) arrives sequentially.
- Customer n:
  - sits at an existing table k with probability proportional to the number of customers already there,
  - starts a new table with probability proportional to α.

Formally:

$$p(z_n = k \mid z_{1:n-1}) = \begin{cases} \frac{n_k}{n-1+\alpha}, & \text{existing table } k, \\ \frac{\alpha}{n-1+\alpha}, & \text{new table.} \end{cases}$$

Partition Interpretation

Let Π_N denote the partition of $\{1, \ldots, N\}$ induced by assignments $z_1, \ldots, z_N$.
Example:

$$z_1 = z_2 = z_7 = z_8 = 1, \quad z_3 = z_5 = z_6 = 2, \quad z_4 = 3$$

$$\Rightarrow \Pi_8 = \{\{1, 2, 7, 8\}, \{3, 5, 6\}, \{4\}\}.$$

A partition consists of mutually exclusive and exhaustive subsets of $\{1, \ldots, N\}$.

Probability of a Seating Arrangement

Let K_N be the number of tables and let $C \in \Pi_N$ denote a cluster.
Then:

$$\mathbb{P}(\Pi_N = \pi_N) = \frac{\alpha^{K_N} \prod_{C \in \Pi_N} (|C| - 1)!}{\alpha(\alpha + 1) \cdots (\alpha + N - 1)}.$$

Exchangeability

The probability of a partition does not depend on the order of customers:

$$\mathbb{P}(\Pi_N = \pi_N) \text{ is invariant under permutations.}$$

Thus, the CRP is *exchangeable*.

Conditional Assignment Rule

Given $\Pi_{N,-n}$ (partition without customer n):

$$p(\Pi_N \mid \Pi_{N,-n}) = \begin{cases} \frac{|C|}{\alpha+N-1}, & \text{if } n \text{ joins cluster } C, \\ \frac{\alpha}{\alpha+N-1}, & \text{if } n \text{ creates new cluster.} \end{cases}$$

Representations of the Dirichlet Process

The Dirichlet process admits several equivalent representations:

- *Posterior form:*

$$G \sim DP(\alpha, H), \quad \theta \mid G \sim G$$
$$\iff \theta \sim H, \quad G \mid \theta \sim DP\left(\alpha + 1, \frac{\alpha H + \delta_\theta}{\alpha + 1}\right)$$

- *Pólya urn:*

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1}$$

- *CRP:*

$$p(z_n = k \mid z_{1:n-1}) = \begin{cases} \frac{n_k}{n-1+\alpha}, \\ \frac{\alpha}{n-1+\alpha} \end{cases}$$

- *Stick-breaking:*

$$\pi_k = \beta_k \prod_{i=1}^{k-1} (1 - \beta_i), \quad \beta_k \sim \text{Beta}(1, \alpha)$$

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*}$$

Exchangeability and Joint Distribution

From the predictive rule:

$$\theta_n \mid \theta_{1:n-1} \sim \frac{\alpha H + \sum_{i=1}^{n-1} \delta_{\theta_i}}{\alpha + n - 1},$$

the joint distribution is:

$$p(\theta_1, \ldots, \theta_n) = \frac{\alpha^K \prod_{k=1}^K h(\theta_k^*)(m_k - 1)!}{\prod_{i=1}^n (\alpha + i - 1)},$$

where:

- θ_k^* are distinct values,
- m_k are their multiplicities,
- $h(\theta)$ is the density under H.

This distribution is exchangeable.
By De Finetti's theorem, this implies the existence of a random measure G such that

$$\theta_1, \theta_2, \ldots \overset{iid}{\sim} G, \quad G \sim DP(\alpha, H).$$

2.12 Number of Clusters

Let K_n denote the number of clusters among n observations.

2.12.1 Expected Value

Define indicator:

$$I_i = \mathbf{1}\{\text{new cluster at step } i\}.$$

Then:

$$\mathbb{E}[K_n] = \sum_{i=1}^n \mathbb{E}[I_i] = \sum_{i=1}^n \frac{\alpha}{\alpha + i - 1}.$$

Thus:

$$\mathbb{E}[K_n] = O(\alpha \log n).$$

2.12.2 Variance

$$\mathrm{Var}(K_n) = \sum_{i=1}^n \frac{\alpha(i-1)}{(\alpha + i - 1)^2}.$$

2.12.3 Distribution of K_n

The distribution satisfies:

$$\mathbb{P}(K_n = k) = \frac{\alpha}{\alpha + n - 1}\mathbb{P}(K_{n-1} = k - 1) + \frac{n-1}{\alpha + n - 1}\mathbb{P}(K_{n-1} = k).$$

Boundary conditions:

$$\mathbb{P}(K_1 = 1) = 1.$$

Closed form:

$$\mathbb{P}(K_n = k) = C_n(k)\, \alpha^k \frac{\Gamma(\alpha)}{\Gamma(\alpha + n)},$$

where $C_n(k)$ are unsigned Stirling numbers of the first kind.

2.12.4 Interpretation

- Rich-get-richer effect: larger clusters attract more points
- Few large clusters, many small ones
- Number of clusters grows sublinearly in n

Chapter 3

Sampling Techniques

3.1 Sampling Techniques for DP Mixture Models

Gaussian Likelihood Setup

Assume:

$$y \sim \mathcal{N}(\mu, \sigma^2), \quad p(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y-\mu)^2\right).$$

For i.i.d. data $y = (y_1, \ldots, y_n)$:

$$L(\mu, \sigma^2) \propto \left(\frac{1}{\sigma^2}\right)^{n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_i (y_i - \mu)^2\right).$$

We focus on:

$$p(\mu, \sigma^2 \mid y) = p(\mu \mid \sigma^2, y)\, p(\sigma^2 \mid y).$$

Semi-Conjugate Prior

Assume independent priors:

$$\pi(\mu, \sigma^2) = \pi(\mu)\pi(\sigma^2),$$

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad \sigma^2 \sim IG\left(\frac{\nu_0}{2}, \frac{\delta_0}{2}\right).$$

Then:

$$\mu \mid \sigma^2, y \sim \mathcal{N}(\mu_n, \tau_n^2),$$

$$\mu_n = \frac{\mu_0/\sigma_0^2 + \bar{y}\, n/\sigma^2}{1/\sigma_0^2 + n/\sigma^2}, \quad \tau_n^2 = \frac{1}{1/\sigma_0^2 + n/\sigma^2}.$$

$$\sigma^2 \mid y \text{ not in closed form}, \quad \sigma^2 \mid y, \mu \sim IG.$$

3.1.1 Finite Mixture Models

A flexible density:

$$f(y) = \sum_{k=1}^{K} \tau_k f(y \mid \theta_k),$$

where:

$$\tau_k \in (0,1), \quad \sum_k \tau_k = 1.$$

3.1.2 Bayesian Mixture Model

$$y_i \mid \tau, \theta \sim \sum_{k=1}^{K} \tau_k f(y_i \mid \theta_k),$$

$$\tau \sim \mathrm{Dir}(\alpha, \dots, \alpha), \quad \theta_k \sim \pi_k(\theta_k), \quad K \sim \pi(K).$$

3.1.3 Gibbs Sampling (Finite Case)

For fixed K:

- Assignment:
$$P(z_i^{(t)} = k \mid \cdot) \propto \tau_k^{(t-1)} f(y_i \mid \theta_k^{(t-1)}).$$
- Weights:
$$\tau \mid z \sim \mathrm{Dir}(\alpha_1 + \hat{n}_1, \dots, \alpha_K + \hat{n}_K),$$
where $\hat{n}_k = \sum_i \mathbf{1}(z_i = k)$.
- Parameters:
$$p(\theta_k \mid \cdot) \propto \pi_k(\theta_k) \prod_{i: z_i = k} f(y_i \mid \theta_k).$$

3.2 CRP Mixture Model

Generative Model

$$\Pi_N \sim CRP(N, \alpha),$$

$$\forall C \in \Pi_N, \quad \phi_C \overset{i.i.d.}{\sim} G_0,$$

$$x_n \mid \phi_C \sim F(\phi_C), \quad n \in C.$$

Inference Goal

$$p(\Pi_N \mid x_{1:N}).$$

Gibbs Sampler (Collapsed)

$$p(\Pi_N \mid \Pi_{N,-n}, x) = \begin{cases} \frac{|C|}{\alpha+N-1}\, p(x_{C\cup\{n\}} \mid x_C), & \text{join } C, \\ \frac{\alpha}{\alpha+N-1}\, p(x_n), & \text{new cluster.} \end{cases}$$

For Gaussian case:

$$p(x_{C\cup\{n\}} \mid x_C) = \mathcal{N}(\tilde{m}, \tilde{\Sigma} + \Sigma),$$

$$\tilde{\Sigma}^{-1} = \Sigma_0^{-1} + |C|\Sigma^{-1}, \quad \tilde{m} = \tilde{\Sigma}\left(\Sigma^{-1} \sum_{m\in C} x_m + \Sigma_0^{-1}\mu_0\right).$$

3.3 DP Mixture Model (DPM)

$$y_i \sim \int p(y_i \mid \theta) G(d\theta), \quad G \sim DP(M, G_0).$$

Example:

$$y_i \sim \int \mathcal{N}(y_i \mid \mu, \sigma^2) G(d\mu).$$

3.4 Alternative Representations

3.4.1 Stick-Breaking Form

$$y_i \mid (w_h, \tilde{\theta}_h) \sim \sum_{h=1}^{\infty} w_h p(y_i \mid \tilde{\theta}_h),$$

$$\tilde{\theta}_h \sim G_0, \quad w_h = v_h \prod_{k<h} (1 - v_k), \quad v_h \sim \text{Beta}(1, M).$$

3.4.2 Clustering Representation

$$y_i \mid \theta_i \sim p(y_i \mid \theta_i), \quad \theta_i \mid G \sim G, \quad G \sim DP(M, G_0).$$

Integrating out G:

$$(\theta_1, \ldots, \theta_n) \sim \text{CRP-induced distribution.}$$

Predictive:

$$p(\theta_{n+1} \mid \theta_{1:n}) \propto \sum_{j=1}^{k_n} n_j \delta_{\theta_j^*} + MG_0.$$

3.5 Collapsed Gibbs Sampling

3.5.1 Conditional for Parameters

$$\theta_i \mid \theta_{-i}, y \propto \sum_{j=1}^{k^-} n_j^- p(y_i \mid \theta_j^*)\delta_{\theta_j^*} + Mp(y_i \mid \theta_i)G_0(\theta_i).$$

Posterior:

$$p(\theta_i \mid y_i, G_0) = \frac{p(y_i \mid \theta_i)G_0(\theta_i)}{\int p(y_i \mid \theta)G_0(d\theta)}.$$

3.5.2 Cluster Assignment Update

$$p(s_i = j \mid s^-, y) \propto \begin{cases} n_j^- \int p(y_i \mid \theta_j^*)dp(\theta_j^* \mid y_j^-), & j \leq k^-, \\ M \int p(y_i \mid \theta)dG_0(\theta), & j = k^- + 1. \end{cases}$$

3.5.3 Cluster Parameter Update

$$p(\theta_j^* \mid s, y) \propto G_0(\theta_j^*) \prod_{i:s_i=j} p(y_i \mid \theta_j^*).$$

3.6 Gaussian DPM Example

$$p(y_i \mid \theta_i) = \mathcal{N}(\theta_i, \sigma^2), \quad G_0 = \mathcal{N}(m, B).$$

$$\int p(y_i \mid \theta)dG_0(\theta) = \mathcal{N}(y_i \mid m, B + \sigma^2).$$

Cluster predictive:

$$\int p(y_i \mid \theta_j^*)dp(\theta_j^* \mid y_j^-) = \mathcal{N}(y_i \mid m_j^-, V_j^- + \sigma^2),$$

$$V_j^- = \left(\frac{1}{B} + \frac{n_j^-}{\sigma^2}\right)^{-1}, \quad m_j^- = V_j^- \left(\frac{m}{B} + \frac{1}{\sigma^2}\sum_{h \in S_j^-} y_h\right).$$

Posterior:

$$p(\theta_j^* \mid y) = \mathcal{N}(m_j, V_j).$$

3.7 Reading: Mixture of Dirichlet Process Models

Model Representation

The Dirichlet process mixture model is

$$y_i \mid \theta_i \sim F(\theta_i), \quad \theta_i \sim G, \quad G \sim DP(\alpha, G_0)$$

Integrating out G gives

$$\theta_i \mid \theta_{1:i-1} = \sum_{j<i} \frac{1}{i-1+\alpha}\delta_{\theta_j} + \frac{\alpha}{i-1+\alpha}G_0$$

This induces clustering through repeated values of θ_i.

Partition View

Let c_i denote cluster assignments. Then

$$P(c_i = c \mid c_{-i}) = \begin{cases} \frac{n_{-i,c}}{n-1+\alpha} \\ \frac{\alpha}{n-1+\alpha} \end{cases}$$

and

$$y_i \mid c_i = c \sim F(\theta_c)$$

Thus inference reduces to learning

$$(\Pi_n, \{\theta_c\})$$

Non-Conjugate Difficulty

Key terms required in Gibbs sampling are

$$\int F(y_i \mid \theta)\, dG_0(\theta) \quad \text{and} \quad p(\theta \mid y_i) \propto F(y_i \mid \theta)G_0(\theta)$$

These are not available in closed form in general.

No-Gaps Representation

Clusters are indexed as $1, \ldots, k$ with no empty labels. This avoids redundant representations of the same partition and simplifies computation.

Augmented Sampling Idea

Introduce auxiliary parameters

$$\theta_1^*, \ldots, \theta_m^* \sim G_0$$

Then approximate the new cluster term via

$$\alpha \int F(y_i \mid \theta)\, dG_0(\theta) \approx \frac{\alpha}{m} \sum_{j=1}^{m} F(y_i \mid \theta_j^*)$$

Gibbs Updates

Cluster assignments are updated via

$$P(c_i = c \mid \cdot) \propto \begin{cases} n_{-i,c} F(y_i \mid \theta_c) \\ \frac{\alpha}{m} F(y_i \mid \theta_c^*) \end{cases}$$

Cluster parameters are updated via

$$\theta_c \mid y_c \sim p(\theta_c \mid y_c)$$

Main Insight

Non-conjugacy can be handled by expanding the state space and performing Gibbs updates in the augmented model, avoiding direct evaluation of intractable integrals.

3.8 Reading: Sampling Methods for Dirichlet Process Mixtures

Dirichlet Process Mixture Representation

The model is

$$y_i \mid \theta_i \sim F(\theta_i), \quad \theta_i \sim G, \quad G \sim DP(\alpha, G_0)$$

which implies

$$\theta_i \mid \theta_{1:i-1} = \sum_{j<i} \frac{1}{i-1+\alpha} \delta_{\theta_j} + \frac{\alpha}{i-1+\alpha} G_0$$

This is equivalent to a Chinese restaurant process on cluster assignments.

Collapsed Gibbs Sampling

If conjugate, integrate out θ_c:

$$P(c_i = c \mid c_{-i}, y) \propto n_{-i,c} \cdot p(y_i \mid y_{c,-i})$$

For a new cluster:

$$\propto \alpha \int F(y_i \mid \theta)\, dG_0(\theta)$$

Failure Under Non-Conjugacy

The integral

$$\int F(y_i \mid \theta)\, dG_0(\theta)$$

is not tractable, so collapsed Gibbs cannot be used.

Metropolis-Hastings Update

Propose

$$\theta^* \sim G_0$$

Accept with probability

$$\rho = \min\left(1, \frac{F(y_i \mid \theta^*)}{F(y_i \mid \theta_i)}\right)$$

The prior cancels, so no integral is needed.

Auxiliary Gibbs Sampling

Introduce auxiliary draws

$$\theta_1^*, \ldots, \theta_m^* \sim G_0$$

Update assignments via

$$P(c_i = c \mid \cdot) \propto \begin{cases} n_{-i,c} F(y_i \mid \theta_c) \\ \frac{\alpha}{m} F(y_i \mid \theta_c^*) \end{cases}$$

Mixing Behavior

Standard Gibbs updates one observation at a time:

$$c_i \to c_i'$$

This leads to slow exploration of partitions and difficulty in splitting or merging clusters.

Conceptual Summary

Posterior inference is over

$$p(\Pi_n, \{\theta_c\} \mid y)$$

Key challenges are

$$\text{non-conjugacy} \quad \text{and} \quad \text{infinite mixture structure}$$

Efficient sampling requires combining

$$\text{Gibbs}, \quad \text{Metropolis-Hastings}, \quad \text{augmentation}$$

3.9 Sampling Techniques: Slice Sampling for DP Mixtures

Hyperpriors for Base Measure

Assume the base measure is parameterized as $\phi = (m, B)$ with priors

$$m \sim \mathcal{N}(m_0, D), \quad B \sim IGamma(a, b)$$

Given cluster parameters $\{\theta_j^*\}_{j=1}^k$, the posterior for m is

$$m \mid \cdots \sim \mathcal{N}(m_1, D_1), \quad D_1^{-1} = \frac{1}{D} + \frac{k}{B}, \quad m_1 = D_1 \left(\frac{m_0}{D} + \frac{1}{B} \sum_{j=1}^k \theta_j^* \right)$$

The posterior for B is

$$B \mid \cdots \sim IGamma(a_1, b_1), \quad a_1 = a + \frac{k}{2}, \quad b_1 = b + \frac{1}{2} \sum_{j=1}^k (\theta_j^* - m)^2$$

Sampling the Concentration Parameter

We use the marginal:

$$p(k \mid M) \propto M^k \frac{\Gamma(M)}{\Gamma(M+n)}$$

Introduce auxiliary variable:

$$\eta \mid M, k \sim Beta(M+1, n)$$

If prior $M \sim Gamma(c, d)$, then:

$$M \mid \eta, k \sim \pi \cdot Gamma(c+k, d - \log \eta) + (1-\pi) \cdot Gamma(c+k-1, d - \log \eta)$$

where mixing weight depends on (k, n, η).

No-Gaps Algorithm

Clusters are labeled consecutively:

$$c_i \in \{1, \ldots, k\}$$

with relabeling applied whenever clusters become empty.
For each observation i, remove y_i and define:

$$k^- = \#\text{ clusters without } i, \quad n_j^- = \sum_{\ell \neq i} \mathbf{1}(c_\ell = j)$$

If y_i is not a singleton:

$$P(c_i = j \mid \cdot) \propto \begin{cases} n_j^- p(y_i \mid \theta_j^*) & j \leq k^- \\ \frac{M}{k^-+1} p(y_i \mid \theta_{k^-+1}^*) & j = k^- + 1 \end{cases}$$

If y_i is a singleton:

$$P(\text{keep } c_i) = \frac{k^- - 1}{k^-}, \quad P(\text{resample}) = \frac{1}{k^-}$$

Updating Cluster Parameters

Given assignments $\{c_i\}$:

$$p(\theta_j^* \mid y) \propto G_0(\theta_j^*) \prod_{i:c_i=j} p(y_i \mid \theta_j^*)$$

If non-conjugate, sample via Metropolis-Hastings.
New cluster candidates:

$$\theta_{k+1}^* \sim G_0$$

Slice Sampling Idea

Target density:

$$\pi(x) = \frac{f(x)}{Z}, \quad Z = \int f(s)\, ds$$

Introduce auxiliary variable u:

$$u \sim Uniform(0, f(x))$$

Define joint:

$$p(x, u) = \frac{1}{Z}\mathbf{1}(0 < u < f(x))$$

Marginalizing:

$$p(x) = \int_0^{f(x)} \frac{1}{Z} du = \frac{f(x)}{Z}$$

Slice Sampling Updates

Sample:

$$u_t \sim Uniform(0, f(x_t))$$

Define slice:

$$S(u) = \{x : f(x) \geq u\}$$

Then update:

$$x_{t+1} \sim Uniform(S(u_t))$$

Slice Sampling for DP Mixtures

Likelihood:

$$p(y_i \mid \{w_h\}, \{\tilde{\theta}_h\}) = \sum_{h=1}^{\infty} w_h p(y_i \mid \tilde{\theta}_h)$$

Introduce slice variable:

$$u_i \sim Uniform(0, w_{z_i})$$

Augmented likelihood:

$$p(y_i, u_i) = \sum_{h=1}^{\infty} \mathbf{1}(u_i < w_h) p(y_i \mid \tilde{\theta}_h)$$

Key property:

$$u_i > 0 \Rightarrow \text{only finitely many } w_h > u_i$$

Thus:

$$\sum_{h=1}^{\infty} \to \sum_{h: w_h > u_i}$$

which is finite.

Marginalization Check

$$\int p(y_i, u_i) du_i = \sum_{h=1}^{\infty} p(y_i \mid \tilde{\theta}_h) \int_0^{w_h} du_i = \sum_{h=1}^{\infty} w_h p(y_i \mid \tilde{\theta}_h)$$

Thus the original model is recovered.

Key Insight

Slice sampling converts:

$$\text{infinite mixture} \quad \to \quad \text{finite active components per iteration}$$

by introducing auxiliary variables $\{u_i\}$.
This enables exact inference without truncation.

Slice Sampler: Component Indicators

Introduce latent indicators $r_i \in \{1, 2, \dots\}$ such that

$$r_i = h \quad \iff \quad y_i \text{ is generated from } \tilde{\theta}_h$$

The augmented likelihood becomes

$$p(y_i, u_i, r_i \mid \{w_h\}, \{\tilde{\theta}_h\}) = \mathbf{1}(u_i < w_{r_i})\, p(y_i \mid \tilde{\theta}_{r_i})$$

Marginalizing out r_i:

$$\sum_{h=1}^{\infty} \mathbf{1}(u_i < w_h) p(y_i \mid \tilde{\theta}_h)$$

recovers the slice-augmented likelihood.

The full joint:

$$p(y, u, r \mid \{w_h\}, \{\tilde{\theta}_h\}) = \prod_{i=1}^{n} \mathbf{1}(u_i < w_{r_i}) p(y_i \mid \tilde{\theta}_{r_i})$$

Slice Sampler: Conditional Updates

Slice variables:

$$u_i \mid r_i, \{w_h\} \sim Uniform(0, w_{r_i})$$

Component indicators:

$$P(r_i = h \mid u_i, y_i, \{\tilde{\theta}_h\}) \propto \mathbf{1}(w_h > u_i)\, p(y_i \mid \tilde{\theta}_h)$$

Only components with $w_h > u_i$ are considered.

Component parameters:

$$p(\tilde{\theta}_h \mid \{y_i : r_i = h\}) \propto G_0(\tilde{\theta}_h) \prod_{i:r_i=h} p(y_i \mid \tilde{\theta}_h)$$

Stick-breaking weights:

Let

$$w_h = v_h \prod_{k<h} (1 - v_k)$$

Then

$$v_h \mid \{r_i\} \sim Beta\left(1 + n_h,\ M + \sum_{k>h} n_k\right), \quad n_h = \sum_i \mathbf{1}(r_i = h)$$

Finite Approximation to DP

Instead of infinite mixtures:

$$G = \sum_{h=1}^{\infty} w_h \delta_{\tilde{\theta}_h}$$

approximate with finite truncation:

$$G_H = \sum_{h=1}^{H} w_h \delta_{\tilde{\theta}_h}$$

The marginal distributions are close:

$$\|p_\infty - p_H\|_1 \leq 4ne^{-(H-1)/M}$$

Thus small H can suffice when M is moderate.

Finite DP Posterior Structure

Joint posterior:

$$p(\{r_i\}, \{\tilde{\theta}_h\}, \{v_h\} \mid y) \propto \prod_{i=1}^{n} p(y_i \mid \tilde{\theta}_{r_i}) \prod_{i=1}^{n} p(r_i \mid \{v_h\}) \prod_{h=1}^{H} dG_0(\tilde{\theta}_h) \prod_{h=1}^{H-1} p(v_h \mid M)$$

Conditionals:

$$p(\tilde{\theta}_h \mid \cdot) \propto G_0(\tilde{\theta}_h) \prod_{i:r_i=h} p(y_i \mid \tilde{\theta}_h)$$

$$v_h \mid \cdot \sim Beta\left(1 + n_h,\ M + \sum_{k>h} n_k\right)$$

$$P(r_i = h \mid \cdot) \propto w_h p(y_i \mid \tilde{\theta}_h)$$

Key Comparison

- Slice sampler: exact, dynamically truncates via $\{u_i\}$
- Finite DP: fixed truncation level H
- Both reduce infinite mixture to finite computation per iteration

3.9.1 Reading: Gibbs Sampling for Stick-Breaking Priors

Stick-Breaking Random Measures

A general stick-breaking prior can be written as

$$G = \sum_{k=1}^{\infty} p_k \delta_{Z_k}$$

where

$$p_1 = V_1, \quad p_k = V_k \prod_{l<k} (1 - V_l), \quad V_k \sim Beta(a_k, b_k)$$

The atoms satisfy

$$Z_k \overset{iid}{\sim} H$$

This defines a broad class of random probability measures including DP and Pitman–Yor.

Finite vs Infinite Construction

Finite case:

$$\sum_{k=1}^{N} p_k = 1 \quad \text{with } V_N = 1$$

Infinite case requires

$$\sum_{k=1}^{\infty} \mathbb{E}[\log(1 - V_k)] = -\infty$$

to ensure weights sum to 1 almost surely.

Dirichlet Process Special Case

DP corresponds to

$$V_k \sim Beta(1, \alpha)$$

giving

$$G \sim DP(\alpha, H)$$

This provides a constructive definition equivalent to Ferguson's original formulation.

Generalized Gibbs Sampling

Two main approaches:

Pólya urn Gibbs:

$$\theta_{n+1} \mid \theta_{1:n} \sim \frac{\alpha}{\alpha + n} H + \sum_k \frac{n_k}{\alpha + n} \delta_{\theta_k^*}$$

Blocked Gibbs: Directly sample

$$\{V_k\}, \{Z_k\}, \{p_k\}$$

This avoids marginalization and improves mixing.

Pitman–Yor Extension

Stick-breaking form:

$$V_k \sim Beta(1 - a, b + ka)$$

This yields richer clustering behavior than DP.

Key Insight

Stick-breaking provides:

- explicit construction of random measures
- direct sampling schemes
- flexibility beyond Dirichlet process

3.10 Reading: Slice Sampling Mixture Models

Dirichlet Process Mixture Model

The model:

$$f(y) = \int K(y \mid \phi)\, dP(\phi), \quad P \sim DP(M, P_0)$$

Stick-breaking form:

$$P = \sum_{j=1}^{\infty} w_j \delta_{\phi_j}, \quad w_j = z_j \prod_{l<j} (1 - z_l), \quad z_j \sim Beta(1, M)$$

Slice Sampling Idea

Introduce auxiliary variable:

$$u_i \sim Uniform(0, w_{d_i})$$

Joint:

$$p(y_i, u_i) = \sum_{j=1}^{\infty} \mathbf{1}(u_i < w_j) K(y_i \mid \phi_j)$$

This restricts to finite set:

$$A_u = \{j : w_j > u_i\}$$

Latent Component Indicator

Introduce

$$d_i \in A_u$$

Then

$$p(y_i, u_i, d_i) = \mathbf{1}(u_i < w_{d_i}) K(y_i \mid \phi_{d_i})$$

This converts infinite mixture into finite conditional problem.

Gibbs Updates

$$\phi_j \mid \cdot \propto p_0(\phi_j) \prod_{i:d_i=j} K(y_i \mid \phi_j)$$

$$z_j \sim Beta\left(1 + \sum_i \mathbf{1}(d_i = j),\ M + \sum_i \mathbf{1}(d_i > j)\right)$$

$$u_i \sim Uniform(0, w_{d_i})$$

$$P(d_i = k) \propto \mathbf{1}(w_k > u_i) K(y_i \mid \phi_k)$$

Key Insight

Slice variables ensure:

- only finitely many components active per iteration
- exact inference without truncation
- easy extension to general stick-breaking priors

3.11 Reading: Bayesian Density Estimation via DP Mixtures

Model Setup

Data:

$$y_i \mid \theta_i \sim N(\mu_i, V_i)$$

Prior:

$$\theta_i \sim G, \quad G \sim DP(\alpha G_0)$$

Clustering Property

Conditional distribution:

$$\theta_i \mid \theta_{-i} \sim \frac{\alpha}{\alpha+n-1} G_0 + \sum_{j \neq i} \frac{1}{\alpha+n-1} \delta_{\theta_j}$$

Leads to clustering (ties among θ_i).

Posterior Predictive

$$y_{n+1} \mid y_{1:n} \sim \frac{\alpha}{\alpha+n} \int K(y \mid \theta) dG_0(\theta) + \sum_k \frac{n_k}{\alpha+n} K(y \mid \theta_k^*)$$

Interpretation

- mixture of new component (prior) and existing clusters
- adaptive smoothing depending on data
- connects to kernel density estimation

Gibbs Sampling Strategy

Update each θ_i:

$$p(\theta_i \mid \theta_{-i}, y) \propto \alpha p(y_i \mid \theta_i) G_0(\theta_i) + \sum_{j \neq i} p(y_i \mid \theta_j) \delta_{\theta_j}$$

This forms basis for CRP-based samplers.

Key Insight

DP mixtures provide:

- flexible density estimation
- automatic clustering
- uncertainty quantification on number of components

3.12 Advanced MCMC Methods

Metropolis–Hastings Algorithm

Given current state $x^{(t)}$:

$$Y_t \sim q(y \mid x^{(t)})$$

Accept with probability

$$\rho(x, y) = \min\left\{\frac{\pi(y)\,q(x \mid y)}{\pi(x)\,q(y \mid x)}, 1\right\}$$

Update:

$$x^{(t+1)} = \begin{cases} Y_t & \text{with prob } \rho(x^{(t)}, Y_t) \\ x^{(t)} & \text{otherwise} \end{cases}$$

3.12.1 Random Walk Metropolis–Hastings (RWMH)

Proposal:

$$Y_t = x^{(t)} + Z_t, \quad Z_t \sim g$$

Thus

$$q(y \mid x) = g(y - x)$$

Common choices:

$$Z_t \sim N(0, \sigma^2 I_d)$$

Scaling of the Proposal

Tradeoff:

$$\sigma \text{ small} \Rightarrow \text{slow exploration}$$

$$\sigma \text{ large} \Rightarrow \text{high rejection}$$

Goal: balance via acceptance rate.

Optimal Acceptance Rates

For high dimension d:

$$\text{Optimal rate} \approx 0.234$$

For $d = 1$:

$$\text{Optimal rate} \approx 0.44$$

Problems with MH

Local Trap

Example:

$$\pi(x) = \frac{1}{3}N(\mu_1, \Sigma) + \frac{2}{3}N(\mu_2, \Sigma)$$

Chains may remain in one mode:

$$P(\text{jump between modes}) \approx \text{small}$$

Intractable Normalizing Constants

Target:

$$f(x) \propto c(x)\psi(x)$$

Acceptance ratio requires:

$$\frac{c(y)}{c(x)}$$

If $c(x)$ unknown $\Rightarrow$ MH not feasible.

3.12.2 Metropolis Adjusted Langevin Algorithm (MALA)

Continuous-time:

$$dX_t = dB_t + \frac{1}{2}\nabla \log \pi(X_t)$$

Discrete proposal:

$$Y_t \sim N\left(x^{(t)} + \frac{\sigma^2}{2}\nabla \log \pi(x^{(t)}), \sigma^2 I_d\right)$$

Properties of MALA

$$\text{Optimal acceptance rate} \approx 0.574$$

Advantages:

$$\text{Uses gradient information} \Rightarrow \text{faster mixing}$$

Limitation:

$$\nabla \log \pi(x) \text{ must be computable}$$

Implementation Issues

Convergence diagnostics:

$$\text{Need full exploration of support}$$

Autocorrelation:

$$\text{Low autocorrelation} \Rightarrow \text{efficient sampling}$$

Multiple Chains vs Single Chain

Multiple chains:

$$\{x_t^{(m)}\}_{m=1}^M$$

Single chain:

$$x_1, x_2, \dots$$

Goal:

$$\text{independent samples} \approx \text{low autocorrelation}$$

Thinning

Keep every k-th sample:

$$\{x_k, x_{2k}, \dots\}$$

Reparametrization

Problem:

$$\text{High posterior correlation}$$

Example:

$$y_{ij} = \mu + \alpha_i + e_{ij}$$

$$\alpha_i \sim N(0, \tau^2), \quad e_{ij} \sim N(0, \sigma^2)$$

Posterior correlations:

$$\mathrm{Cor}(\mu, \alpha_i) \approx -\left(1 + \frac{\sigma^2/n}{\tau^2/m}\right)^{-1/2}$$

Reparametrize:

$$\beta_i = \mu + \alpha_i$$

Leads to reduced correlation and improved mixing.

Blocking

Group parameters:

$$\theta = (\theta_1, \dots, \theta_k)$$

Instead of scalar updates:

$$\theta_j \mid \theta_{-j}$$

Use block updates:

$$(\theta_{i_1}, \dots, \theta_{i_k}) \mid \theta_{-(i_1,\dots,i_k)}$$

Benefit:

$$\text{Better exploration in correlated directions}$$

Collapsing

Marginalize variables:

$$p(x, y) \to p(x)$$

Example:

$$(X_1, X_2, X_3) \to (X_1, X_2)$$

Then sample:

$$X_3 \mid X_1, X_2$$

Benefit:

Lower variance, faster convergence

Auxiliary Variable Methods

Introduce u:

$$p(x) = \int p(x, u)\, du$$

Goal:

simplify sampling or avoid intractable terms

Examples:

Slice sampling

Key principle:

augment state space to simplify conditionals

Key Takeaways

Efficiency depends on proposal design and parameterization

Use gradients (MALA), blocking, collapsing, or augmentation

Avoid local traps and high autocorrelation

3.13 Auxiliary Variable Methods and Hybrid MCMC

Cauchy–Normal Model via Data Augmentation

$$X_i \overset{iid}{\sim} \text{Cauchy}(\mu, 1)$$

Likelihood:

$$L(\mu \mid x_{1:n}) = \prod_{i=1}^{n} \frac{1}{\pi(1 + (x_i - \mu)^2)}$$

Prior:

$$\mu \sim N(0, 10)$$

Posterior:

$$\pi(\mu \mid x) \propto e^{-\mu^2/20} \prod_{i=1}^{n} \frac{1}{1 + (x_i - \mu)^2}$$

Introduce latent variables:

$$\frac{1}{1 + (x_i - \mu)^2} = \int_0^\infty e^{-\omega_i(1+(x_i-\mu)^2)} d\omega_i$$

Augmented posterior:

$$\pi(\mu, \omega \mid x) \propto e^{-\mu^2/20} \prod_{i=1}^{n} e^{-\omega_i(1+(x_i-\mu)^2)}$$

Gibbs updates:

$$\omega_i \mid \mu \sim \mathrm{Exp}(1 + (x_i - \mu)^2)$$

$$\mu \mid \omega \sim N\left(\frac{\sum_i \omega_i x_i}{\sum_i \omega_i + 1/20}, \frac{1}{2\sum_i \omega_i + 1/10}\right)$$

Metropolis-within-Gibbs

When a full conditional is not tractable:

$$\theta_j \mid \theta_{-j} \quad \text{sampled via MH}$$

Procedure:

$$Y \sim q(\cdot \mid \theta_j)$$

$$\text{accept with prob } \min\left\{\frac{\pi(Y \mid \theta_{-j})q(\theta_j \mid Y)}{\pi(\theta_j \mid \theta_{-j})q(Y \mid \theta_j)}, 1\right\}$$

Key idea:

Replace difficult Gibbs steps with MH steps

3.13.1 Adaptive Gibbs Samplers

Goal:

improve efficiency by learning during sampling

Selection probabilities:

$$\alpha_n = (\alpha_{n,1}, \ldots, \alpha_{n,d})$$

Update rule:

$$\alpha_n = R_n(\alpha_{n-1}, X_{n-1}, \ldots)$$

Transition:

$$X_n \sim P_{\alpha_n}(X_{n-1}, \cdot)$$

Key condition (diminishing adaptation):

$$|\alpha_n - \alpha_{n-1}| \to 0$$

Ensures:

ergodicity under suitable conditions

3.13.2 Random Scan Gibbs Sampler

State:

$$X = (X_1, \ldots, X_d)$$

At each step:

$$i \sim \text{Categorical}(\alpha_1, \ldots, \alpha_d)$$

$$X_i \sim \pi(\cdot \mid X_{-i})$$

Kernel:

$$P_\alpha = \sum_{i=1}^{d} \alpha_i P_i$$

Adaptive Random Scan Gibbs

Update:

$$\alpha_n = R_n(\alpha_{n-1}, X_{0:n-1})$$

Then:

$$X_n \sim P_{\alpha_n}(X_{n-1}, \cdot)$$

Key result:

$$|\alpha_n - \alpha_{n-1}| \to 0 \quad \Rightarrow \quad \text{ergodicity (under conditions)}$$

Metropolis-within-Gibbs (Adaptive)

If conditional unavailable:

$$X_i \sim Q_{x_{-i}}(X_i, \cdot)$$

Accept:

$$\min\left\{1, \frac{\pi(Y \mid x_{-i})q(Y, X_i)}{\pi(X_i \mid x_{-i})q(X_i, Y)}\right\}$$

Adaptive version:

$$\gamma_n \text{ controls proposal}$$

$$Q_n = Q_{\gamma_n}$$

Condition:

$$\|Q_{n+1} - Q_n\| \to 0$$

Ergodicity Conditions (Key Structure)

Two critical requirements:

Diminishing adaptation:

$$\|P_n - P_{n-1}\| \to 0$$

Containment:

$$\text{chain does not escape to infinity}$$

Together imply:

$$X_n \xrightarrow{d} \pi$$

Failure of Naive Adaptation

Even if:

$$\alpha_n \to \alpha$$

Chain may still be:

$$\text{non-ergodic (transient)}$$

Hence:

$$\text{adaptation must be controlled carefully}$$

3.13.3 Adaptive Metropolis-within-Gibbs (Full)

Updates both:

$$\alpha_n \quad \text{and} \quad \gamma_n$$

Algorithm:

$$\alpha_n = R_n(\cdot), \quad \gamma_n = R'_n(\cdot)$$

Then:

$$X_n \sim P_{\alpha_n,\gamma_n}(X_{n-1}, \cdot)$$

Slice Sampling (Revisited)

Joint:

$$p(x,u) = \frac{1}{Z}\mathbf{1}(0 < u < f(x))$$

Gibbs steps:

$$u \mid x \sim \text{Unif}(0, f(x))$$

$$x \mid u \sim \text{Unif}(\{x : f(x) \geq u\})$$

For DP mixtures:

$$u_i < w_{r_i} \Rightarrow \text{finite truncation}$$

Key Insight Across Methods

All methods aim to:

$$\text{improve mixing}$$

via:

$$\text{better proposals}, \quad \text{reparameterization}, \quad \text{augmentation}$$

Unifying principle:

$$\text{modify the Markov kernel while preserving } \pi$$

Takeaways

$$\text{Gibbs: simple but may mix slowly}$$

$$\text{MH: flexible but sensitive to tuning}$$

$$\text{MALA: uses gradients}$$

$$\text{Adaptive MCMC: learns structure, but requires care}$$

$$\text{Auxiliary variables: turn hard problems into easy conditionals}$$

3.14 Pitman–Yor Process (PYP)

Motivation: Limitations of the Dirichlet Process

- DP mixture model:
  - standard Bayesian nonparametric clustering model
  - exchangeable partitions
  - flexible number of clusters
- Key limitations:
  - cluster weights decay exponentially
  - number of clusters grows slowly:

$$\mathbb{E}[K_n] \sim \alpha \log n$$

- Consequence:
  - underestimates many small clusters
  - not suitable for heavy-tailed cluster structures

Idea of the Pitman–Yor Process

- Goal:
  - minimal extension of DP
  - preserve exchangeability
  - maintain tractable inference
- Key modification:
  - introduce discount parameter d
  - decouple:
    - reinforcement (rich-get-richer)
    - tail behavior of cluster sizes
- Result:
  - exponential decay $\to$ power-law decay
  - heavier tails in cluster sizes

Definition: Stick-Breaking Representation

- Random measure:
$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k}, \quad \phi_k \overset{iid}{\sim} G_0$$
- Weights:
$$\pi_k = v_k \prod_{\ell<k}(1 - v_\ell)$$
- Stick-breaking variables:
$$v_k \sim \mathrm{Beta}(1 - d, \alpha + kd)$$
- Parameter constraints:
$$0 \le d < 1, \quad \alpha > -d$$
- Special case:
$$d = 0 \Rightarrow \text{Dirichlet Process}$$

Pitman–Yor via Discounted CRP

- Let:
  - K_n = number of clusters
  - n_k = size of cluster k
- Assignment probabilities:
$$\mathbb{P}(z_{n+1} = k \mid z_{1:n}) = \begin{cases} \frac{n_k - d}{n+\alpha}, & k = 1, \dots, K_n \\ \frac{\alpha + dK_n}{n+\alpha}, & \text{new cluster} \end{cases}$$

- Interpretation:
  - reduced reinforcement: $n_k - d$ instead of n_k
  - increased innovation: $\alpha + dK_n$

Key Properties

- Discounted reinforcement:
  - weakens "rich-get-richer"
  - small clusters survive longer
- Cluster-dependent innovation:
  - more clusters $\Rightarrow$ higher probability of new cluster
- Balance:
  - prevents dominance of large clusters
  - encourages diversity

Power-Law Behavior of Weights

- Expected weight:
$$\mathbb{E}[\pi_k] = \mathbb{E}[v_k] \prod_{\ell<k} \mathbb{E}[1 - v_\ell]$$
- Moments:
$$\mathbb{E}[v_k] = \frac{1-d}{\alpha + kd + 1 - d}$$
$$\mathbb{E}[1 - v_\ell] = \frac{\alpha + \ell d}{\alpha + \ell d + 1 - d}$$
- Asymptotics:
$$\mathbb{E}[\pi_k] \sim C_{\alpha,d}\, k^{-1/d}$$
- Interpretation:
  - polynomial decay (heavy tail)
  - larger $d \Rightarrow$ heavier tail

Growth of Number of Clusters

- New cluster probability:
$$\mathbb{P}(z_{n+1} = \text{new}) = \frac{\alpha + dK_n}{n + \alpha}$$
- Let $m_n = \mathbb{E}[K_n]$, then:
$$m_{n+1} = m_n + \frac{\alpha + dm_n}{n + \alpha}$$

- Asymptotic approximation:
$$\frac{dm_n}{dn} \approx \frac{dm_n}{n}$$
- Solve:
$$\log m_n \approx d \log n \quad \Rightarrow \quad m_n \sim Cn^d$$
- Compare:
  - DP: $\mathbb{E}[K_n] \sim \log n$
  - PYP: $\mathbb{E}[K_n] \sim n^d$

Takeaways

- PYP generalizes DP with one extra parameter d
- introduces power-law cluster sizes
- supports many small clusters
- better fits real-world heavy-tailed data
- retains tractable CRP-style inference

Chapter 4

Extensions of the Dirichlet Process

4.1 Pitman–Yor Process: Inference and Applications

4.1.1 Inference under Pitman–Yor Process

- Model:
$$G \sim \mathrm{PYP}(\alpha, d, G_0), \quad \theta_i \mid G \sim G, \quad y_i \mid \theta_i \sim F(\theta_i)$$
- Integrating out G yields a collapsed Gibbs sampler similar to DP mixtures
- Cluster assignment update:
$$\mathbb{P}(z_i = k \mid z_{-i}, y) \propto \begin{cases} (n_{k,-i} - d)\, p(y_i \mid y_{k,-i}), & k \le K_{-i} \\ (\alpha + dK_{-i})\, p(y_i \mid G_0), & k = \text{new} \end{cases}$$
- Same structure as DP mixture:
  - only prior weights differ
  - likelihood terms unchanged
- Computational cost:
  - comparable to DP inference

DP vs PYP: Key Differences

- Parameters:
$$\text{DP: } \alpha > 0, \quad \text{PYP: } \alpha > -d,\ 0 \le d < 1$$
- Stick-breaking:
$$\text{DP: } v_k \sim \mathrm{Beta}(1, \alpha), \quad \text{PYP: } v_k \sim \mathrm{Beta}(1-d, \alpha + kd)$$
- Weight decay:
$$\text{DP: exponential}, \quad \text{PYP: power-law}$$

- Cluster growth:

$$\mathbb{E}[K_n] \sim \log n \quad \text{(DP)}, \qquad \mathbb{E}[K_n] \sim n^d \quad \text{(PYP)}$$

- Reinforcement:

$$\text{DP: } n_k, \quad \text{PYP: } n_k - d$$

- New cluster probability:

$$\text{DP: } \frac{\alpha}{n+\alpha}, \quad \text{PYP: } \frac{\alpha + dK_n}{n+\alpha}$$

When to Use Pitman–Yor

- DP assumptions:
  - new clusters quickly become unlikely
  - tail mass decays exponentially
  - small clusters disappear
- Use PYP when:
  - persistent rare categories
  - heavy-tailed cluster sizes
  - continual discovery of new types
- Applications:
  - genetics (rare alleles)
  - text modeling (long-tail words)
  - ecology (species abundance)
  - social networks (degree distributions)

4.1.2 Case Study: Genetics

- Setup:
  - n individuals, each with allele at a locus
  - K_n distinct alleles, counts n_k
- Empirical pattern:
  - few common alleles
  - many rare alleles
- Model:

$$G \sim \text{PYP}(\alpha, d, G_0), \quad z_i \mid G \sim G$$

- Benefit:
  - heavy-tailed cluster sizes preserve rare alleles

4.1.3 Case Study: Topic Modeling

- Topic distributions:
$$\phi_k = (\phi_{k,1}, \phi_{k,2}, \dots), \quad \sum_v \phi_{k,v} = 1$$
- Word generation:
$$w_i \mid z_i = k \sim \text{Categorical}(\phi_k)$$
- Empirical observation:
  - few very frequent words
  - many rare words
- Model:
$$\phi_k \sim \text{PYP}(\alpha, d, G_0)$$
- Effect:
  - heavy-tailed word distributions
  - preserves rare words

4.2 Hierarchical Dirichlet Process (HDP)

Motivation: Multiple Related Groups

- Data partitioned into groups:
$$\{x_{ji}\}, \quad j = 1, \dots, J, \; i = 1, \dots, n_j$$
- Within each group:
  - observations are exchangeable
  - each group has its own mixture distribution
- Goal:
  - unknown number of components
  - components shared across groups
  - group-specific weights

Why Sharing Matters

- Unknown number of clusters:
  - parametric models require fixing K
  - nonparametric models allow growth with data

- Shared components:
  - groups are not independent
  - large groups inform small ones
- Different weights:
  - prevalence varies across groups

Failure of Naive Construction

- Attempt:
$$G_j \sim \mathrm{DP}(\alpha_j, G_0(\tau)), \quad \tau \sim p(\tau)$$
- Conditional independence:
$$G_j \perp G_{j'} \mid \tau$$
- Problem:
  - if $G_0(\tau)$ is continuous:
$$\mathbb{P}(\theta_{jk} = \theta_{j'k'}) = 0$$
  - no shared atoms
- Conclusion:
  - sharing parameter τ is insufficient
  - need discrete base measure

HDP Model Definition

- Global measure:
$$G_0 \mid \gamma, H \sim \mathrm{DP}(\gamma, H)$$
- Group-specific measures:
$$G_j \mid \alpha_0, G_0 \sim \mathrm{DP}(\alpha_0, G_0)$$
- Observations:
$$\theta_{ji} \mid G_j \sim G_j, \quad x_{ji} \mid \theta_{ji} \sim F(\theta_{ji})$$
- Key property:
  - G_0 is discrete
  - atoms are shared across all G_j

4.3 Hierarchical Dirichlet Process: Constructions and Inference

Graphical Interpretation and Limitation of Naive Model

- Naive construction:
$$G_j \sim \mathrm{DP}(\alpha_j, G_0(\tau)), \quad \tau \sim p(\tau)$$
- Conditional independence:
$$G_j \perp G_{j'} \mid \tau$$
- Even with shared hyperparameter τ, atoms differ across groups:
  - G_j are independent draws
  - no shared support
- Conclusion:
  - no sharing of mixture components
  - motivates hierarchical construction

4.3.1 Stick-Breaking Construction of HDP

- Global stick-breaking:
$$G_0 = \sum_{k=1}^{\infty} \beta_k \delta_{\phi_k}, \quad \phi_k \overset{iid}{\sim} H, \quad \beta \sim \mathrm{GEM}(\gamma)$$
- Group-level measures:
$$G_j \mid G_0 \sim \mathrm{DP}(\alpha_0, G_0)$$
- Since G_0 is discrete, all G_j share atoms $\{\phi_k\}$:
$$G_j = \sum_{k=1}^{\infty} \pi_{jk} \delta_{\phi_k}$$
- Goal:
  - derive stick-breaking form for π_{jk}

Finite-Dimensional Marginals

- For partition $(A_1, \ldots, A_r)$:
$$(G_j(A_1), \ldots, G_j(A_r)) \sim \mathrm{Dir}(\alpha_0 G_0(A_1), \ldots, \alpha_0 G_0(A_r))$$

- Using decomposition:

$$G_j(A_\ell) = \sum_{k \in K_\ell} \pi_{jk}, \quad G_0(A_\ell) = \sum_{k \in K_\ell} \beta_k$$

- Leads to:

$$\left(\sum_{k \in K_1} \pi_{jk}, \ldots, \sum_{k \in K_r} \pi_{jk} \right) \sim \mathrm{Dir}\left(\alpha_0 \sum_{k \in K_1} \beta_k, \ldots \right)$$

Stick-Breaking for Group Weights

- Representation:

$$\pi_{jk} = \pi'_{jk} \prod_{\ell < k} (1 - \pi'_{j\ell})$$

- Where:

$$\pi'_{jk} \sim \mathrm{Beta}\left(\alpha_0 \beta_k, \ \alpha_0 \sum_{\ell > k} \beta_\ell \right)$$

- Derived using:
  - Dirichlet aggregation
  - renormalization properties

4.3.2 Chinese Restaurant Franchise (CRF)

- Hierarchical model:

$$G_0 \sim \mathrm{DP}(\gamma, H), \quad G_j \sim \mathrm{DP}(\alpha_0, G_0)$$

- Interpretation:
  - each group = restaurant
  - observations = customers
  - tables = local clusters
  - dishes = global clusters

CRF Variables

- Global dishes:

$$\phi_k \sim H$$

- Table-level assignments:

$$\psi_{jt} = \phi_{k_{jt}}$$

- Customer-level:

$$\theta_{ji} = \psi_{j,t_{ji}}$$

- Observations:

$$x_{ji} \sim F(\theta_{ji})$$

Predictive Distribution for Customers

- Integrating out G_j:

$$\theta_{ji} \mid \theta_{j,1:i-1} \sim \sum_{t=1}^{m_j} \frac{n_{jt}}{i-1+\alpha_0}\delta_{\psi_{jt}} + \frac{\alpha_0}{i-1+\alpha_0}G_0$$

- Interpretation:
  - join existing table with prob $\propto n_{jt}$
  - start new table with prob $\propto \alpha_0$

Predictive Distribution for Tables

- Integrating out G_0:

$$\psi_{jt} \mid \{\psi\}_{-jt} \sim \sum_{k=1}^{K} \frac{m_k}{m_\cdot+\gamma}\delta_{\phi_k} + \frac{\gamma}{m_\cdot+\gamma}H$$

- Where:
  - m_k = number of tables serving dish k
  - $m_\cdot = \sum_k m_k$
- Interpretation:
  - reuse global dish with prob $\propto m_k$
  - create new dish with prob $\propto \gamma$

Index-Based Sampling View

- Customer $\to$ table:

$$\mathbb{P}(t_{ji} = t \mid \cdot) = \begin{cases} \frac{n_{jt}}{i-1+\alpha_0}, & t \leq m_j \\ \frac{\alpha_0}{i-1+\alpha_0}, & t = m_j + 1 \end{cases}$$

- Table $\to$ dish:

$$\mathbb{P}(k_{jt} = k \mid \cdot) = \begin{cases} \frac{m_k}{m_\cdot+\gamma}, & k \leq K \\ \frac{\gamma}{m_\cdot+\gamma}, & k = K + 1 \end{cases}$$

Collapsed Gibbs Sampling: Table Assignment Update

- For customer i in group j, the table assignment update has the Gibbs form

$$p(t_{ji} = t \mid \text{rest}, x) \propto p(t_{ji} = t \mid \text{rest})\, p(x_{ji} \mid t_{ji} = t, \text{rest})$$

- The prior term comes from the CRP within group j:

$$p(t_{ji} = t \mid t^{-ji}) \propto \begin{cases} n_{jt}^{-ji}, & \text{if table } t \text{ exists,} \\ \alpha_0, & \text{if table } t \text{ is new.} \end{cases}$$

- The likelihood term is collapsed by integrating out the dish parameter ϕ_k.

Collapsed Predictive Density

Define the collapsed predictive density

$$f_k^{-x_{ji}}(x_{ji}) = p(x_{ji} \mid \{x_{j'i'} : z_{j'i'} = k,\ (j', i') \neq (j, i)\}).$$

Using Bayes' rule,

$$f_k^{-x_{ji}}(x_{ji}) = \frac{\int f(x_{ji} \mid \phi_k) \prod_{j'i' \neq ji,\, z_{j'i'}=k} f(x_{j'i'} \mid \phi_k) h(\phi_k)\, d\phi_k}{\int \prod_{j'i' \neq ji,\, z_{j'i'}=k} f(x_{j'i'} \mid \phi_k) h(\phi_k)\, d\phi_k}.$$

- h is the density of the base measure H.
- The denominator is the marginal likelihood of observations currently assigned to global cluster k, excluding x_{ji}.
- The numerator is the marginal likelihood if x_{ji} were added to cluster k.
- If $h(\phi)$ is conjugate to $f(x \mid \phi)$, this predictive density is available in closed form.

Collapsed Gibbs Sampling: Dish Assignment Update

- For a newly created table t in group j, the dish assignment update is

$$p(k_{jt} = k \mid \text{rest}, x) \propto p(k_{jt} = k \mid \text{rest})\, p(x_{jt} \mid k_{jt} = k, \text{rest}).$$

- The prior term comes from the global CRP:

$$p(k_{jt} = k \mid k^{-jt}) \propto \begin{cases} m_{\cdot k}, & \text{if dish } k \text{ exists,} \\ \gamma, & \text{if dish } k \text{ is new.} \end{cases}$$

- The likelihood term is collapsed over ϕ_k. If table t is assigned dish k, all customers at that table share parameter ϕ_k:

$$p(x_{jt} \mid k, \text{rest}) = \prod_{i:t_{ji}=t} f_k^{-x_{ji}}(x_{ji}).$$

- Since the table was just created, it contains the single customer that triggered the new table:

$$\prod_{i:t_{ji}=t} f_k^{-x_{ji}}(x_{ji}) = f_k^{-x_{ji}}(x_{ji}).$$

- Therefore:

$$p(k_{jt} = k \mid \cdot) \propto \begin{cases} m_{\cdot k}\, f_k^{-x_{ji}}(x_{ji}), & \text{existing dish } k, \\ \gamma\, f_{\text{new}}^{-x_{ji}}(x_{ji}), & \text{new dish.} \end{cases}$$

4.3.3 Case Study: BNP Modeling of Marathon Data

- Context:
  - Marathon finishing times depend on age, gender, and race conditions.
  - Traditional grading methods rely on top records and coarse adjustments.
  - Goal: data-driven comparison using latent structure.
- Problem:
  - Discover latent *running patterns* (pace profiles across race segments).
  - Cluster runners within each age/gender group.
  - Share patterns across groups using an HDP.
- Setup:
  - Groups indexed by j (age/gender).
  - Runner i in group j represented by:
  $$x_{ji} = \text{(fraction of time spent in each segment)}.$$
- HDP model:
  - Global distribution:
  $$G_0 \sim \mathrm{DP}(\gamma, H)$$
  - Group-level:
  $$G_j \sim \mathrm{DP}(\alpha, G_0), \quad G_j = \sum_k \pi_{jk} \delta_{\phi_k}$$
  - Likelihood:
  $$x_{ji} \mid z_{ji} = k \sim \mathrm{Dirichlet}(\tau \phi_k)$$
- Interpretation:
  - ϕ_k = global pace pattern
  - π_{jk} = prevalence of pattern in group j
- Data:
  - NYC Marathon (2007–2011)
  - $\sim$ 194,000 runners
- Outcome:
  - $\sim$ 46 latent pace clusters discovered
  - Examples:
    - fast-start then slow-down
    - steady pace
- Insight:
  - Groups differ in mixture weights but share global patterns
  - Enables prediction of finishing time from partial race data

4.4 Dependent Dirichlet Processes (DDP)

Motivation

- Standard DP assumes a single distribution:

$$G \sim \mathrm{DP}(\alpha, H)$$

- In regression:

$$y_i = f(x_i) + \epsilon_i$$

- Need distributions that vary with x:

$$G_x \quad \text{depends on } x$$

DDP Construction

- Define a collection:

$$\mathcal{G} = \{G_x : x \in \mathcal{X}\}$$

- Marginals:

$$G_x \sim \mathrm{DP}(\cdot, \cdot)$$

- Stick-breaking representation:

$$G_x = \sum_h p_h \delta_{m_h(x)}$$

- Key idea:
  - weights p_h may be shared
  - atoms $m_h(x)$ vary with x

Types of Dependence

- Common weights, dependent atoms:

$$p_h \text{ fixed}, \quad m_h(x) \text{ varies with } x$$

- Common atoms, dependent weights
- Both dependent
- Partial sharing via discrete base measure

Example: Health Outcomes

- Variables:
  - Y = health outcome
  - x_1 = exposure
  - $x_2, \ldots, x_p$ = covariates
- Interest:

$$\text{distribution of } Y \mid x$$

- Motivation:
  - effects may appear in tails
  - avoid dichotomizing outcomes

DDE and Preterm Birth Example

- Logistic model:

$$\log \Pr(y_i = 1 \mid x_i, \beta) = \beta_0 + \sum_{h=1}^{4} \beta_h \, dde_{hi} + z_i^\top \alpha$$

- Interpretation:
  - exposure categories modeled via indicators dde_{hi}
  - z_i = confounders
- Results:
  - increasing odds ratios across exposure levels
  - strong evidence of dose-response relationship

Key Insight of DDP

- Instead of modeling only the mean:

$$\mathbb{E}[Y \mid x]$$

- Model full conditional distribution:

$$Y \mid x \sim G_x$$

- Allows:
  - heteroskedasticity
  - multimodality
  - tail effects

Mixture Models and Regression Extensions

Basic Mixture Model

- Let y_i denote gestational age for individual i.
- Model density as a mixture:

$$f(y_i) = \int \mathcal{N}(y_i; \mu, \sigma^2)\, dG(\mu, \sigma^2)$$

- Finite mixture approximation:

$$G = \sum_{h=1}^{k} p_h \delta_{\theta_h}, \quad \theta_h = (\mu_h, \sigma_h^2)$$

- Mixtures of Gaussians can approximate flexible densities.

Mixtures of Regression Models

- Incorporate predictors x_i (e.g., DDE):

$$f(y_i \mid x_i) = \sum_{h=1}^{k} p_h \mathcal{N}(y_i; x_i' \beta_h, \sigma_h^2)$$

- Hierarchical mixture of experts (HME):
  - Replace weights p_h with $p_h(x_i)$
  - Allows predictor-dependent clustering
- Typically estimated via EM:
  - Can overfit
  - Requires choosing k

Infinite Mixtures Motivation

- Avoid fixing number of components k
- Introduce nonparametric prior:

$$G_x = \{G_x : x \in \mathcal{X}\}$$

- If G is global, use:

$$G \sim \mathrm{DP}(\alpha G_0)$$

4.4.1 Dependent Dirichlet Processes (DDP)

General Form

- Define predictor-dependent random measure:

$$G_x = \sum_{h=1}^{\infty} p_h \delta_{\theta_h(x)}$$

- Atoms vary with x:

$$\theta_h(\cdot) \sim \text{Gaussian Process}$$

- Weights may be:
  - shared across x
  - or predictor-dependent

Key Idea

- Introduce dependence across x through:
  - weights $p_h(x)$, or
  - locations $\theta_h(x)$
- Goal: flexible conditional distributions $f(y \mid x)$

4.4.2 Weighted Mixture of Dirichlet Processes

- Construct local DPs at knots x_j^*:

$$G_j^* \sim \text{DP}(M, G_0)$$

- Define:

$$G_x(B) = \sum_{j=1}^{J} \frac{K(x, x_j^*)}{\sum_\ell K(x, x_\ell^*)} G_j^*(B)$$

- Interpretation:
  - Nearby knots receive higher weight
  - Smooth variation across predictor space

4.4.3 Kernel Stick-Breaking Process (KSBP)

Construction

- Define:

$$G_x = \sum_{h=1}^{\infty} \left[V_h K(x, \Gamma_h) \prod_{\ell<h} (1 - V_\ell K(x, \Gamma_\ell)) \right] G_h^*$$

- Components:
  - $V_h \sim \mathrm{Beta}(a_h, b_h)$
  - $\Gamma_h \sim H$ (locations)
  - $G_h^* \sim \mathrm{DP}(\alpha G_0)$
- $K(x, \Gamma_h)$ is a bounded kernel

Interpretation

- Weights depend on distance between x and Γ_h
- Nearby components get larger weights
- Induces smooth dependence across predictors

Special Cases

- If $K(x, \Gamma_h) = 1$:
  - recovers standard DP or Pitman–Yor (depending on V_h)
- If $K(x, \Gamma_h) = \exp(-\psi \|x - \Gamma_h\|)$:
  - weights decay with distance

4.4.4 Predictor-Dependent Clustering

- Marginal predictive:

$$(\phi_i \mid \phi^{(i)}, X) = \frac{\alpha}{\alpha + w_i} G_0 + \sum_{j \neq i} \frac{w_{ij}}{\alpha + w_i} \delta_{\phi_j}$$

- Weights w_{ij} depend on similarity in predictors
- Key difference from DP:
  - clustering depends on x

Hierarchical Dirichlet Process (HDP) Revisited

- Model:

$$G_0 \sim \mathrm{DP}(\gamma, H), \quad G_j \mid G_0 \sim \mathrm{DP}(\alpha, G_0)$$

- Interpretation:
  - G_0 defines global atoms
  - G_j selects subsets with group-specific weights
- Enables sharing across groups without explicit regression

4.4.5 Example: Information Retrieval

- Words:
$$x_i \sim G_j$$
- G_j: document-specific distribution
- G_0: global vocabulary distribution
- Topics shared across documents

4.4.6 Nested Dirichlet Process

- Model:
  - G_j themselves are clustered
  - $G_j \sim Q$, where Q is a DP over distributions
- Two levels:
  - clustering of groups (e.g., states)
  - clustering within groups (e.g., hospitals)
- Allows discovery of higher-level structure

4.4.7 ANOVA Dependent Dirichlet Process (ANOVA-DDP)

Model Construction

- Consider categorical predictors:
$$x_j = (v, w)$$
(e.g., row/column effects in ANOVA)
- Introduce dependence across x via ANOVA structure:
$$m_{xh} = \mu_h + \alpha_{h,v} + \beta_{h,w}$$
- Each mixture component varies with factor levels while sharing global structure.

Equivalent Representation

- Random measure:
$$G_x(\theta) = \sum_h w_h \delta_{m_{xh}}$$
- Can be written as:
$$\theta_{xi} = \Phi_i' d_x$$
- Where:

  - d_x = design vector (ANOVA encoding)
  - $M_h = (\mu_h, \alpha_{1,h}, \ldots, \alpha_{V,h}, \beta_{1,h}, \ldots, \beta_{W,h})$
- Interpretation:
  - ANOVA-DDP = DP mixture of ANOVA models

4.4.8 Problem Setting for BNP Regression

- Data:
$$\{(y_i, x_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^Q$$
- Model:
$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$
- Objective:
  - Learn a distribution over functions $p(f)$

Applications

- Regression (curve fitting, smoothing)
- Classification (categorical outputs)
- Prediction (generalization)

4.4.9 Gaussian Process (GP)

Definition

- A Gaussian process is a distribution over functions:
$$f \sim \mathrm{GP}(m(x), K(x, x'))$$
- Any finite collection $\{f(x_1), \ldots, f(x_n)\}$ is jointly Gaussian.

Covariance Structure

- Covariance matrix:
$$K_{ij} = K(x_i, x_j)$$
- Interpretation:
$$K(x_i, x_j) = \mathbb{E}[f(x_i) f(x_j)]$$
- Must be positive semidefinite.

4.4.10 Common Covariance Functions

Squared Exponential (SE)

- Kernel:

$$K(x, x') = \sigma_0^2 \exp\left(-\frac{1}{2}\left(\frac{x - x'}{\lambda}\right)^2\right)$$

- Properties:
  - Smooth (infinitely differentiable)
  - Stationary
- Hyperparameters:
  - λ: lengthscale
  - σ_0: amplitude

Matérn Class

- Kernel:

$$K(x, x') = \frac{2^{1-\nu}}{\Gamma(\nu)}\left(\frac{\sqrt{2\nu}|x - x'|}{\lambda}\right)^\nu K_\nu\left(\frac{\sqrt{2\nu}|x - x'|}{\lambda}\right)$$

- Controls smoothness:
  - $\nu \to \infty$: SE kernel
  - $\nu = 1/2$: exponential kernel

Other Kernels

- Linear:

$$K(x, x') = \sigma_0^2 + xx'$$

- Brownian motion:

$$K(x, x') = \min(x, x')$$

- Periodic:

$$K(x, x') = \exp\left(-\frac{2\sin^2((x - x')/2)}{\lambda^2}\right)$$

4.4.11 Combining Kernels

- Sum:

$$K = K_1 + K_2$$

- Product:

$$K = K_1 K_2$$

- Convolution:

$$K(x, x') = \int h(x, z) K(z, z') h(x', z')\, dz\, dz'$$

4.4.12 GP Regression

- Prior:
$$f \sim \mathrm{GP}(m, K)$$
- Likelihood:
$$y \mid f \sim \mathcal{N}(f(X), \sigma^2 I)$$
- Posterior:
$$f \mid y \sim \mathrm{GP}(m_p, K_p)$$
- Posterior mean:
$$m_p(x) = m(x) + K(X, x)^T (K + \sigma^2 I)^{-1} (y - m(X))$$
- Posterior covariance:
$$K_p(x, x') = K(x, x') - K(X, x)^T (K + \sigma^2 I)^{-1} K(X, x')$$

4.4.13 Prediction (Multi-step)

- Joint Gaussian:
$$\begin{bmatrix} y \\ y^* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right)$$
- Conditional mean:
$$\mathbb{E}[y^* \mid y] = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(y - \mu_1)$$
- Conditional variance:
$$\mathrm{Var}(y^* \mid y) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

4.4.14 Learning Hyperparameters

- Marginal likelihood:
$$p(y \mid X, \theta) = \mathcal{N}(0, K_\theta + \sigma^2 I)$$
- Log marginal likelihood:
$$\log p(y \mid X, \theta) = -\frac{1}{2} \log |K_\theta + \sigma^2 I| - \frac{1}{2} y^T (K_\theta + \sigma^2 I)^{-1} y + \mathrm{const}$$
- Estimation:
  - Maximum likelihood
  - Fully Bayesian (e.g., HMC)

4.4.15 Remarks

- GPs:
  - handle uncertainty via averaging
  - learn kernel structure from data
  - avoid manual feature engineering
- Compared to SVM:
  - probabilistic (uncertainty quantification)
  - automatic regularization

4.4.16 Feature Selection

- Model selection problem:
$$m \in \{0,1\}^D$$
- Ideal solution:
$$\hat{m} = \arg\max_m p(D \mid m)$$
- Challenges:
  - combinatorial (2^D models)
  - overfitting
- Bayesian solution:
  - average over models instead of selecting one

4.4.17 Feature Selection in Bayesian Models

- Key questions:
  - Why perform feature selection?
  - What is the cost of including all features?
- Important distinction:
  - Overfitting is less relevant in fully Bayesian methods (no point estimation)
  - Bayesian averaging already regularizes models
- Feature selection is justified when:
  - Measurement cost is high
  - Prediction with many features is computationally expensive
- Empirical note:
  - Bayesian methods can perform well using all features

Automatic Relevance Determination (ARD)

- Kernel with dimension-specific lengthscales:

$$K(x_n, x_{n'}) = \nu \exp\left(-\frac{1}{2}\sum_{d=1}^{D}\left(\frac{x_n^{(d)} - x_{n'}^{(d)}}{r_d}\right)^2\right)$$

- Interpretation:
  - r_d controls variation along dimension d
  - $r_d \to \infty \Rightarrow$ dimension becomes irrelevant
- Learning $\{r_d\}_{d=1}^{D}$ enables automatic feature selection.

4.4.18 Limitations of Gaussian Processes

- Computational cost:
$$\mathcal{O}(n^3)$$
due to matrix inversion
- Poor handling of:
  - discontinuities
  - sharp structural changes
- Smooth kernels (e.g., RBF) may be overly restrictive.

4.4.19 Scalable Gaussian Processes

Global Approximations

- Use subset of size $m \ll n$:
  - random subsampling
  - clustering (e.g., centroids)
  - active learning selection
- Low-rank approximation:
$$K_{nn} \approx K_{nm} K_{mm}^{-1} K_{mn}$$

Sparse Kernels

- Compact support kernels:

$$K(x_i, x_j) = 0 \quad \text{if } \|x_i - x_j\| > c$$

- Challenges:

  - maintain positive semidefiniteness

- Complexity:
$$\mathcal{O}(\alpha n^3)$$
with sparsity factor α.

Sparse GP (Inducing Points)

- Introduce $M \ll N$ inducing points
- Approximate prior:
$$p(f) \approx \mathcal{N}(0, K_{NM}K_{MM}^{-1}K_{MN} + \Lambda)$$
- Benefits:
  - Faster inversion:
$$\mathcal{O}(M^2N) \ll \mathcal{O}(N^3)$$
  - Efficient prediction:
$$\mathcal{O}(M) \text{ mean}, \quad \mathcal{O}(M^2) \text{ variance}$$

4.4.20 CART: Regression Trees

- Motivation:
  - Simple global models fail on complex data
- Idea:
  - Partition input space
  - Fit simple models locally
- Implementation:
  - Recursive binary splits
  - Optimize fit subject to complexity constraints

4.4.21 Bayesian Regression Trees

- Model:
$$y(x) = g(x; T, M) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$
- Parameters:
  - T: tree structure
  - M: leaf parameters
- Prior factorization:

$$\pi(T, M, \sigma^2) = \pi(M \mid T, \sigma^2)\pi(T \mid \sigma^2)\pi(\sigma^2)$$

4.4.22 Bayesian Additive Regression Trees (BART)

- Model:
$$y(x) = \sum_{j=1}^{m} g(x; T_j, M_j) + \epsilon$$
- Interpretation:
  - Ensemble of weak learners
  - Each tree captures partial structure
- Prior:
$$\pi((T_1, M_1), \ldots, (T_m, M_m), \sigma^2) = \prod_{j=1}^{m} \pi(M_j \mid T_j, \sigma^2)\pi(T_j \mid \sigma^2)\pi(\sigma^2)$$

4.4.23 Regression Tree Representation

- Tree defines mapping:
$$g(x; T, M) \mapsto \mu(x)$$
- Leaf parameters:
$$M = (\mu_1, \ldots, \mu_b)$$
- Possible leaf models:
  - Constant
  - Linear
  - Gaussian process

4.4.24 MCMC for Tree Models

- Sampling scheme:
  - Sample tree structure T via MH
  - Sample leaf parameters M via Gibbs
  - Sample variance σ^2 via Gibbs

Tree Proposals (MH)

- Grow: split a terminal node
- Prune: collapse a subtree
- Change: modify split rule
- Swap: exchange split rules
- Ensures reversibility of the Markov chain.

4.4.25 Pros and Cons

Advantages

- Highly flexible function approximation
- Handles mixed data types naturally
- Scales via parallel MCMC

Limitations

- Poor mixing (due to discrete tree moves)
- Limited interpretability
- Can underrepresent uncertainty

4.5 Random Partition Models

Random Probability Measures and Clustering

- In Bayesian nonparametrics, we place priors on random distributions:

$$F \sim p(F), \quad F = \sum_{h=1}^{\infty} w_h \delta_{m_h}$$

- Many such priors (e.g., DP) are discrete $\Rightarrow$ induce ties.
- Implied clustering:
  - Draw $\theta_i \sim F$, i.i.d.
  - Positive probability that $\theta_i = \theta_{i'}$
  - This defines clusters via shared values
- Partition:

$$S_j = \{i : \theta_i = \theta_j^*\}, \quad j = 1, \ldots, k$$

- Applications:
  - Text: words grouped into topics
  - Biostatistics: patients grouped into subpopulations

4.5.1 Inference on Partitions

- If F is not of direct interest:

$$\rho = \{S_1, \ldots, S_k\}$$

- Prefer specifying a prior directly on partitions:

$$p(\rho)$$

Notation for Random Partitions

- Units: $i \in S = \{1, \ldots, n\}$
- Partition:
$$\rho_n = \{S_1, \ldots, S_k\}, \quad S = \bigcup_{j=1}^{k} S_j$$
- Cluster sizes:
$$n_j = |S_j|$$
- Allocation representation:
$$s_i = j \iff i \in S_j$$
- Covariate-dependent partitions:
$$p(\rho_n \mid x)$$

4.5.2 Sampling Model

- Conditional on partition:
$$p(y^n \mid \rho, \theta) = \prod_{j=1}^{k} \prod_{i \in S_j} p(y_i \mid \theta_j)$$
- Components:
  - Cluster parameters θ_j
  - Prior on θ_j: often conjugate
  - Prior on ρ: defines clustering structure

4.5.3 Product Partition Model (PPM)

- Partition prior:
$$p(\rho_n) \propto \prod_{j=1}^{k} c(S_j)$$
- $c(S_j)$: cohesion function measuring cluster strength
- Examples:
  - Dirichlet process:
$$c(S_j) = \alpha(n_j - 1)!$$
  - Contiguity (changepoint models):
$$c(S_j) = p^{n_j - 1}(1 - p)\mathbf{1}\{S_j \text{ contiguous}\}$$
- Conjugacy:
  - Posterior remains PPM with updated cohesion

4.5.4 Species Sampling Models (SSM)

- Partition probability depends only on cluster sizes:

$$p(\rho_n) = p(|S_1|, \ldots, |S_k|)$$

- Exchangeable partition probability function (EPPF)
- Predictive characterization (PPF):

$$p(s_{i+1} = j \mid s_1, \ldots, s_i)$$

- Must satisfy consistency conditions for exchangeability

Polya Urn (Dirichlet Process)

- Let k_i be number of clusters among first i observations
- Predictive rule:

$$p(s_{i+1} = \begin{cases} j & \text{existing cluster} \\ k_i + 1 & \text{new cluster} \end{cases}) = \begin{cases} \frac{n_j}{M+i} \\ \frac{M}{M+i} \end{cases}$$

- Equivalent to:
  - DP prior on G
  - Special case of PPM and SSM

Model-Based Clustering

- Mixture model:

$$p(y_i \mid k, \theta, \pi) = \sum_{j=1}^{k} \pi_j f_j(y_i \mid \theta_j)$$

- Latent assignment:

$$p(y_i \mid s_i = j, \theta) = f_j(y_i \mid \theta_j), \quad \Pr(s_i = j) = \pi_j$$

- Induces partition via latent variables s_i

4.5.5 Covariate-Dependent Clustering

Motivation

- Clustering should depend on covariates x_i
- Similar $x_i \Rightarrow$ more likely to co-cluster

4.5.5.1 Example 1: Sarcoma Trial

- 12 cancer subtypes
- Covariate: prognosis category
- Goal: estimate success probabilities

Issues:

- Separate analysis: too little data
- Full pooling: oversmoothing
- Regression: insufficient flexibility

Solution:

- Cluster subtypes
- Share strength within clusters
- Let data determine clustering

4.5.5.2 Example 2: Patient Clustering

- Cluster patients by response trajectories
- Challenge:
  - Want clustering informed by covariates
  - But also predictive for new observations

Augmented Response Approach

- Define augmented response:
$$\tilde{y}_i = (y_i, x_i)$$
- Issues:
  - Complex likelihood construction
  - Incorrect modeling of covariate influence
- Better:
  - Model partition directly via $p(\rho \mid x)$

Mixtures with Covariate-Dependent Weights

- Model:
$$y_i \mid x_i \sim \sum_j \pi_j(x_i) f_j(y_i \mid \theta_j)$$
- Properties:
  - Partition induced via latent s_i

  - Dependence enters through $\pi_j(x)$
- Limitation:
  - Fixed number of components (typically)

Dependent Dirichlet Process (DDP)

- Define predictor-indexed random measures:

$$G_x = \sum_h \pi_h(x)\delta_{\theta_h}$$

- Features:
  - Shared atoms θ_h
  - Predictor-dependent weights $\pi_h(x)$
- Induces covariate-dependent partitions

4.5.6 DDP-like Constructions

Probit Stick-Breaking Process

- Weights:

$$\pi_h(x) = \Phi(\eta_h(x)) \prod_{\ell<h} \{1 - \Phi(\eta_\ell(x))\}$$

- $\eta_h(x)$ controls dependence on covariates

Weighted Mixture of DPs

- Combine local DPs:

$$G_x(B) = \sum_{l=1}^{L} \frac{\gamma_l f(x, x_l^*)}{\sum_{l'} \gamma_{l'} f(x, x_{l'}^*)} G_l^*(B)$$

- Shared atoms, location-dependent weights

Product Partition with Covariates

- Modify cohesion:

$$p(\rho) \propto \prod_{j=1}^{k} c(S_j) \exp\{t \cdot g(S_j)\}$$

- $g(S_j)$ measures similarity within cluster
- Example:

$$g(S_j) = \frac{2}{|S_j|(|S_j| - 1)} \sum_{i<i'} \left(1 - \frac{d_{ii'}}{(1+\epsilon)d^*}\right)$$

- Encourages clustering of nearby points

Summary

- Random partitions provide a direct way to model clustering
- Key frameworks:
  - PPM: cohesion-based
  - SSM: size-based
  - DP: Polya urn
  - Model-based: latent mixtures
- Extensions:
  - Covariate-dependent partitions
  - Dependent random measures
  - Attraction-based clustering

4.5.7 Item Allocation and Attraction-Based Clustering

- Modify predictive probability function (PPF):

$$p(s_{i+1} \mid s_1, \ldots, s_i) = \begin{cases} j & \propto \frac{1}{M+i}\, q(i+1, S_j) \\ k_i + 1 & \propto \frac{M}{M+i} \end{cases}$$

- Attraction function:

$$q(i, S) = \sum_{\ell \in S} a_{i\ell}$$

- Define similarity weights:
  - Distance d_{ij} between items
  - $a_{ij} \propto [(1+\epsilon)d^* - d_{ij}]^T$
  - Normalize: $\sum_{j \neq i} a_{ij} = n - 1$
- Result:
  - Encourages clustering of nearby points
  - Induces ergodic Markov chain (irreducible, aperiodic, recurrent)

4.5.8 PPM with Covariates (PPMx)

- Goal:

$$p(\rho) \to p(\rho \mid x)$$

- Partition prior:

$$p(\rho_n \mid x^n) \propto \prod_{j=1}^{k} g(x_j^*)\, c(S_j)$$

- $g(x_j^*)$: similarity function for cluster S_j
- Construction:

$$g(x_j^*) = \int \prod_{i \in S_j} q(x_i \mid \xi_j) q(\xi_j) \, d\xi_j$$

- Interpretation:
  - Measures within-cluster covariate homogeneity
  - Penalizes heterogeneous clusters

Properties of PPMx

- Symmetry:

$$p(\rho_n \mid x^*) \text{ invariant to permutations}$$

- Coherence across n:

$$p(\rho_n \mid x^n) = \sum_{s_{n+1}} \int p(\rho_{n+1} \mid x^{n+1}) q(x_{n+1} \mid x^n) \, dx_{n+1}$$

- Requires:

$$q(x_{n+1} \mid x^n) = \frac{g_{n+1}(x^{n+1})}{g_n(x^n)}$$

Choice of Similarity Models

- Continuous covariates:

$$q(x_i \mid \xi_j) = \mathcal{N}(\xi_j, V)$$

- Categorical:
  - Multinomial likelihood
  - Dirichlet prior
- Ordinal:
  - Multinomial probit models
- Count data:

$$q(x_i \mid \xi_j) = \text{Poisson}(\xi_j), \quad q(\xi_j) = \text{Gamma}(a, b)$$

Posterior Inference under PPMx

- Augmented model:

$$q(y^n, x^n, \theta, \xi, \rho_n) = \prod_j \prod_{i \in S_j} p(y_i \mid \theta_j, x_i) q(x_i \mid \xi_j) \, p(\theta_j) \, q(\xi_j) \, c(S_j)$$

- Posterior:

$$p(\rho_n \mid x^n, y^n) \propto p(y^n, x^n \mid \rho_n)$$

- Computation:
  - Reduces to standard clustering after augmentation
  - Straightforward Gibbs updates
- Warning:
  - Joint modeling of (x, y) is not equivalent to PPMx

4.5.9 Example: Mixed Effects Model

- Data:
  - $n = 47$ patients
  - repeated measures y_{ij}
  - covariate x_i (dose)
- Model:
  - Cluster patients
  - Share parameters within clusters
- Sampling:
$$p(y_i \mid \theta_j^*, \rho_n) = \mathcal{N}(\theta_j^*, \Sigma)$$
- Partition prior:
$$p(\rho_n \mid x^n) \propto \prod_j c(S_j) g(x_j^*)$$
- Similarity:
  - Multivariate normal model

Covariate-Based vs Augmented Response

- Augmented approach:
  - Treat (x, y) jointly
  - Clustering influenced by both
- Covariate-based clustering:
  - Partition depends only on x
  - Response modeled conditionally
- Key difference:
  - Augmented models distort similarity structure
  - PPMx preserves interpretability of clustering

Chapter 5

Latent Feature Models and the Indian Buffet Process (IBP)

5.1 Infinite Limit Construction

Key Idea: Used vs Unused Features

- Partition columns into:
  - Used: $m_k > 0$
  - Unused: $m_k = 0$
- Define:
$$K_+ = \text{number of used features}$$
- Reorder columns so:
$$m_k > 0 \text{ for } k \leq K_+, \quad m_k = 0 \text{ for } k > K_+$$
- Factorization:
  - Product over unused columns repeats
  - Product over used columns carries structure

Algebraic Simplification

- Using Gamma identities:
$$\frac{\Gamma(N + 1 + \alpha/K)}{\Gamma(1 + \alpha/K)} = \prod_{j=1}^{N} \left(j + \frac{\alpha}{K}\right)$$
- And:
$$\frac{\Gamma(m_k + \alpha/K)}{\Gamma(\alpha/K)} = \left(\frac{\alpha}{K}\right) \prod_{j=1}^{m_k-1} \left(j + \frac{\alpha}{K}\right)$$

Limiting Distribution ($K \to \infty$)

- Harmonic number:

$$H_N = \sum_{j=1}^{N} \frac{1}{j}$$

- Limit:

$$P([Z]) = \frac{\alpha^{K_+}}{\prod_h K_h!} e^{-\alpha H_N} \prod_{k=1}^{K_+} \frac{(N - m_k)!(m_k - 1)!}{N!}$$

- Interpretation:
  - α^{K_+} rewards new features
  - $e^{-\alpha H_N}$ controls total growth
  - Exchangeable distribution

5.2 Indian Buffet Process (IBP)

- Infinite buffet interpretation:
- Customer 1:

$$K_1 \sim \text{Poisson}(\alpha)$$

- Customer i:
  - Chooses existing feature k with probability:

$$\frac{m_k}{i}$$

  - Draws new features:

$$\text{Poisson}\left(\frac{\alpha}{i}\right)$$

- Result:
  - Infinite features available
  - Finite number active per observation

5.2.1 Posterior Inference

- Finite model:

$$P(z_{ik} = 1 \mid Z_{-i,k}, X) \propto P(z_{ik} = 1 \mid Z_{-i,k}) P(X \mid Z)$$

- Beta-Bernoulli marginal:

$$P(z_{ik} = 1 \mid Z_{-i,k}) = \frac{m_{-i,k} + \alpha/K}{N + \alpha/K}$$

- Infinite limit:
$$P(z_{ik} = 1 \mid Z_{-i,k}) = \frac{m_{-i,k}}{N}$$
- New features:
$$\text{Poisson}(\alpha/N)$$

5.2.2 Properties of IBP

- Exchangeable over observations
- Row sums:
$$\#\text{features per observation} \sim \text{Poisson}(\alpha)$$
- Total number of nonzeros:
$$\sim \text{Poisson}(N\alpha)$$
- Number of active features:
$$K_+ \sim \text{Poisson}(\alpha H_N)$$

5.3 Two-Parameter Extension

- Extension introduces β:
$$\pi_k \sim \text{Beta}\left(\frac{\alpha\beta}{K}, \beta\right)$$
- Predictive rule:
  - Existing features:
$$\frac{m_k}{\beta + i - 1}$$
  - New features:
$$\text{Poisson}\left(\frac{\alpha\beta}{\beta + i - 1}\right)$$
- Effect:
  - β controls sparsity and feature reuse

Beta Process Connection

- Beta process:
$$B \sim \text{BP}(c, B_0)$$
- Generates weights:
$$B = \sum p_i \delta_{\omega_i}$$

- Bernoulli process:

$$X \sim \mathrm{BeP}(B)$$

- Interpretation:
  - Beta process generates feature probabilities
  - Bernoulli process generates binary matrix Z

Stick-Breaking Construction for Beta Process

- Generate:

$$\mu_k \sim \mathrm{Beta}(\alpha, 1)$$

- Define weights:

$$\pi_k = \prod_{j=1}^{k} \mu_j$$

- Differences from DP:
  - Weights do NOT sum to 1
  - Features are independent

Conjugacy

- Model:

$$B \sim \mathrm{BP}(c, B_0), \quad X_i \sim \mathrm{BeP}(B)$$

- Posterior:

$$B \mid X_{1:n} \sim \mathrm{BP}\left(c+n, \frac{c}{c+n} B_0 + \frac{1}{c+n} \sum_{i=1}^{n} X_i\right)$$

5.4 IBP from the Beta–Bernoulli Process

- Let:

$$B \sim \mathrm{BP}(c, B_0), \quad X_i \mid B \sim \mathrm{BeP}(B)$$

- Predictive distribution:

$$X_{n+1} \mid X_{1:n} \sim \mathrm{BeP}\left(\frac{c}{c+n} B_0 + \frac{1}{c+n} \sum_{i=1}^{n} X_i\right)$$

- Interpretation:
  - Existing feature j selected with probability:

$$\frac{m_{n,j}}{c+n}$$

  - New features:

$$\text{Poisson}\left(\frac{c\gamma}{c+n}\right)$$

- This recovers the IBP predictive rule.

5.5 Hierarchical Beta Process

- Model:

$$B \sim \text{BP}(c, B_0), \quad A_j \sim \text{BP}(c', B), \quad X_{ij} \sim \text{BeP}(A_j)$$

- Interpretation:
  - Global feature pool: B
  - Group-specific features: A_j
  - Observations draw from group-level features
- Enables:
  - Sharing features across groups
  - Group-specific variability

5.6 Completely Random Measures (CRM)

- A random measure μ assigns random mass:

$$\mu(A) \geq 0, \quad A \subset \Theta$$

- Completely random if:

$$A_1 \cap A_2 = \emptyset \Rightarrow \mu(A_1) \perp \mu(A_2)$$

5.6.1 Lévy–Khintchine Representation

- Any CRM can be written as:

$$\mu = \sum_{k=1}^{\infty} w_k \delta_{\theta_k}$$

- (w_k, θ_k) arise from a Poisson point process with Lévy measure:

$$\nu(dw, d\theta)$$

5.7 Poisson Point Process

- Number of points in region A:

$$N(A) \sim \text{Poisson}(\nu(A))$$

- Independence:

$$A_1 \cap A_2 = \emptyset \Rightarrow N(A_1) \perp N(A_2)$$

- Conditional distribution:
  - Given $N(A) = k$, points are i.i.d. in A

5.8 Gamma Process

- Definition:

$$\mu \sim \text{GaP}(cH, b)$$

- Lévy measure:

$$\nu(dw, d\theta) = cw^{-1}e^{-bw}dw\, H(d\theta)$$

- Key idea:
  - Produces positive weights
  - Infinite number of atoms

5.8.1 Connection to Dirichlet Process

- Normalize:

$$G = \frac{\mu}{\mu(\Theta)}$$

- Then:

$$G \sim \text{DP}(\alpha, H)$$

- Interpretation:
  - Gamma process → unnormalized measure
  - Normalization → probability measure

5.9 Beta Process

- Definition:

$$B \sim \text{BP}(c, H)$$

- Lévy measure:

$$\nu(dw, d\theta) = cw^{-1}(1-w)^{c-1}dw\, H(d\theta)$$

- Properties:
  - $w_k \in (0, 1)$
  - Interpreted as feature probabilities

Beta Process → Bernoulli Process → IBP

- Given:

$$B = \sum_{k=1}^{\infty} w_k \delta_{\theta_k}$$

- Generate features:

$$z_{ik} \mid B \sim \text{Bernoulli}(w_k)$$

- Feature vector:

$$X_i = \sum_{k} z_{ik} \delta_{\theta_k}$$

- Marginalizing B:
  - induces IBP over Z

5.10 Interpretation Across Processes

- Gamma process:
  - weights on $(0, \infty)$
  - normalized → Dirichlet process
- Beta process:
  - weights on $(0, 1)$
  - used as feature probabilities
- Bernoulli process:
  - generates binary features
- IBP:
  - marginal distribution over binary matrix Z

Posterior Inference for IBP Models

- Conditional update:

$$P(z_{ik} = 1 \mid Z_{-i,k}, X) \propto \frac{m_{-i,k}}{N} \cdot P(X \mid Z)$$

- New features:

$$\text{Poisson}(\alpha/N)$$

- Structure:
  - Columns independent
  - Rows exchangeable

5.11 Example: Linear Gaussian Latent Feature Model

- Data:

$$x_i \in \mathbb{R}^d$$

- Model:

$$x_i \sim \mathcal{N}(z_i A, \sigma^2 I)$$

- Where:
  - z_i binary feature vector
  - A feature loading matrix
- Matrix form:

$$X = ZA + \epsilon$$

- Remains well-defined as $K \to \infty$

Key Properties of IBP Models

- Infinite-dimensional feature space
- Finite features per observation
- Exchangeable rows
- Flexible latent representation

Extension: Feature Modeling across multiple groups

Suppose we collect articles from multiple outlets and each article may contain multiple latent topics (politics, tech, sports). The goal is to automatically discover

- What topics exist
- Which topics appear in each article
- how topic usage differs across news sources

We do this by using the Beta Process and IBP framework

$$B \sim BP(c, H), \quad B = \sum_{k=1}^{\infty} w_k \delta_{\theta_k}$$

Where θ_k is the latent topic and w_k is the probability that the topic appears. Each article independently chooses which topics it uses based on

$$z_{ik} \sim \text{Bernoulli}(w_k)$$

where $z_{ik} = 1$ if article i uses topic k and 0 otherwise.

Limitation of the Vanilla Beta Process

Under the standard model

$$P(z_{ik} = 1) = w_k \quad \forall i$$

All articles share the same topic probabilities, which implies that CNN articles use the same topic frequencies as BBC articles, and this is unrealistic as different news outlets ephasize different topics. We want a model with a *shared global pool of topics* but a group-specific topic probability:

- *Global feature measure* defines a shared collection of latent topics

$$B_0 \sim BP(c_0, H) \quad B_0 = \sum_{k=1}^{\infty} p_k \delta_{\theta_k}$$

- *Group specific feature measures*: Each news source j gets its own topic probabilities

$$B_j | B_0 \sim BP(c_1, B_0) \quad B_j = \sum_{k=1}^{\infty} w_{jk} \delta_{\theta_k}, w_{jk} \sim \text{Beta}(c_1 p_k, c_1(1 - p_k))$$

- *Data Generation*: set of topics appearing in article i from news source j

$$X_{ji} | B_j \sim BeP(B_j) \quad z_{jik} | B_j \sim \text{Bernoulli}(B_j)$$

5.12 From IBP to Count-Valued Feature Models

- Recall Beta–Bernoulli construction:

$$B \sim \text{BP}(c, H), \quad X_i \mid B \sim \text{BeP}(B)$$

- Produces binary feature usage:

$$z_{ik} \in \{0, 1\}$$

- Limitation:
  - Many real datasets are count-valued (not binary)

- Example:
  - Gene expression (RNA-seq)
  - Counts of mRNA transcripts per cell
- Motivation:
  - Need models that extend IBP to count data

5.13 Negative Binomial Factor Analysis (NBFA)

- Data:

$$X_{vj} = \text{count of gene } v \text{ in cell } j$$

- Model:

$$X_{vj} \sim \mathrm{NB}\left(\sum_k \phi_{vk}\theta_{kj},\ p_j\right)$$

- Components:
  - ϕ_{vk}: gene loading for factor k
  - θ_{kj}: activity of factor k in sample j
  - p_j: dispersion parameter
- Interpretation:
  - Counts arise from multiple latent factors
- Augmentation:

$$X_{vj} = \sum_k X_{vjk}, \quad X_{vjk} \sim \mathrm{NB}(\phi_{vk}\theta_{kj}, p_j)$$

5.14 Double Feature Allocation (DFA)

- Goal:
  - Model relationships between two sets (e.g., patients and symptoms)
- Idea:
  - Allow overlapping clusters in both dimensions
- Application:
  - Electronic health records
  - Phenotyping

5.15 Latent Disease Interpretation

- Latent disease:
  - Unobserved condition generating symptoms
- Structure:
  - Patient–disease relationships
  - Disease–symptom relationships
- Observed data:
  - Patient symptom matrix

5.16 Patient–Disease (PD) Model

- Feature allocation via IBP:

$$A = (\alpha_{ik}) \in \{0,1\}^{n\times\infty} \sim \mathrm{IBP}(m)$$

- Interpretation:
  - Column k = latent disease
  - $\alpha_{ik} = 1$ means patient i has disease k
- Properties:
  - Finite number of active diseases for finite n

5.17 Symptom–Disease (SD) Model

- Conditional on PD model:

$$C = (\gamma_{lk}) \in \{-1,0,1\}^{p\times K}$$

- Prior:

$$C \sim \mathrm{Categorical}(\pi), \quad \pi \sim \mathrm{Dirichlet}(\phi)$$

- Interpretation:

$$\gamma_{lk} = \begin{cases} -1 & \text{disease } k \text{ decreases symptom } l \\ 0 & \text{no relation} \\ +1 & \text{disease } k \text{ increases symptom } l \end{cases}$$

5.18 Sampling Model

- Data:

$$y_{il} \in \{-1, 0, +1\}$$

- Model:

$$y_{il} \sim \text{Categorical}\left(Me^{\alpha_i^\top W^- \gamma_l^- + \eta_l^-},\ M,\ Me^{\alpha_i^\top W^+ \gamma_l^+ + \eta_l^+}\right)$$

- Components:
  - W^-, W^+: weight matrices
  - η^-, η^+: noise terms
  - M: normalization constant

5.19 Scalability Challenges

- MCMC inference:
  - Sequential updates of A
  - Slow convergence
- Large n:
  - Many rows (patients)
  - Many features (columns)
  - High memory cost

5.20 Consensus Monte Carlo

- Idea:
  1. Split data into subsets
  2. Run MCMC independently
  3. Combine posterior samples
- Limitation for IBP:
  - Feature labels are not aligned across subsets
  - Cannot average discrete feature matrices

5.21 Modified Consensus Monte Carlo

- Fix:
  1. Introduce overlapping subsets (anchors)

2. Align features across subsets
3. Combine via anchor comparison

- Result:
  - More scalable IBP inference

Chapter 6

Polya Tree Priors

6.1 Prior Construction

- So far we have introduced:
  - Dirichlet process (DP): prior on random probability measures for clustering
  - Indian buffet process (IBP): prior on binary latent feature matrices
- Alternative approach: construct priors via recursive partitioning (Polya tree)
- Intuition:
  - A histogram approximates a distribution via bins
  - Fixed partitions are too rigid
  - Polya tree recursively refines partitions

6.2 Ingredients of a Polya Tree

A Polya tree requires:

- Recursive partition of the sample space
- Random branching probabilities at each split
- Path rule defining bin probabilities

6.3 Recursive Partition Tree

- Let $(\Omega, \mathcal{B})$ be a measurable space
- Define a sequence of partitions:

$$\Pi = \{\pi_m : m = 0, 1, 2, \dots\}$$

- Properties:

  - $\pi_0 = \{\Omega\}$
  - π_{m+1} refines π_m
  - Each set splits into two disjoint children

6.3.1 Binary Indexing

- Let:
$$E = \{0,1\}, \quad E^m = \underbrace{E \times \cdots \times E}_{m \text{ times}}, \quad E^* = \bigcup_{m=0}^{\infty} E^m$$
- Each node indexed by binary string $\epsilon \in E^m$
- Partition:
$$\pi_m = \{B_\epsilon : \epsilon \in E^m\}$$
- Tree structure:
$$B_\epsilon = B_{\epsilon 0} \cup B_{\epsilon 1}, \quad B_{\epsilon 0} \cap B_{\epsilon 1} = \emptyset$$

6.4 Polya Tree Definition

- Define branching probabilities:
$$Y_\epsilon := G(B_{\epsilon 0} \mid B_\epsilon)$$
- Then:
$$G(B_{\epsilon 0}) = G(B_\epsilon) Y_\epsilon, \quad G(B_{\epsilon 1}) = G(B_\epsilon)(1 - Y_\epsilon)$$
- Prior:
$$Y_\epsilon \sim \text{Beta}(\alpha_{\epsilon 0}, \alpha_{\epsilon 1})$$
- Probability of node:
$$G(B_\epsilon) = \prod_{j=1}^{m} Y_{\epsilon_1 \cdots \epsilon_{j-1}}^{1-\epsilon_j} (1 - Y_{\epsilon_1 \cdots \epsilon_{j-1}})^{\epsilon_j}$$

6.5 Tailfree Processes

- Define conditional probabilities:
$$Y_\epsilon = \frac{G(B_{\epsilon 0})}{G(B_\epsilon)}$$
- A prior is **tailfree** if:
  - collections at each level are independent:
$$\{Y_\emptyset\}, \{Y_0, Y_1\}, \{Y_{00}, Y_{01}, Y_{10}, Y_{11}\}, \ldots$$
- Polya tree is tailfree

6.6 Dirichlet Process as a Special Case

- Suppose:
$$Y_\epsilon \sim \mathrm{Beta}(\alpha_{\epsilon 0}, \alpha_{\epsilon 1})$$
- With:
$$\alpha_\epsilon = \alpha_{\epsilon 0} + \alpha_{\epsilon 1}, \quad \alpha_\epsilon = c\, H(B_\epsilon)$$
- Then:
$$G \sim \mathrm{DP}(cH)$$

6.6.1 Consistency Property

- If:
$$\alpha_0 = \alpha_{00} + \alpha_{01}, \quad \alpha_1 = \alpha_{10} + \alpha_{11}$$
- Then:
$$(G(B_{00}), G(B_{01}), G(B_{10}), G(B_{11})) \sim \mathrm{Dirichlet}(\alpha_{00}, \alpha_{01}, \alpha_{10}, \alpha_{11})$$
- By induction:
$$(G(B_\epsilon) : |\epsilon| = m) \sim \mathrm{Dirichlet}(\alpha_\epsilon : |\epsilon| = m)$$

6.7 Interpretation

- Polya tree = distribution over distributions
- Built via recursive Beta splits
- Provides more local control than DP
- Can encode prior beliefs about regions of Ω

6.8 Absolute Continuity of Polya Trees

- Absolute continuity result (Kraft, 1964):
  If the concentration parameters grow quadratically,
$$\alpha_{\epsilon_1 \cdots \epsilon_m} = cm^2,$$
  then the Polya tree measure G is absolutely continuous with probability one.
- Intuition:
  At level m,
$$Y_\epsilon \sim \mathrm{Beta}(\alpha_m, \alpha_m), \quad \alpha_m = cm^2$$
  - Mean: $1/2$

  - Variance: $\mathcal{O}(1/\alpha_m)$

- As $m \to \infty$:
  - Splits become increasingly balanced
  - Mass divides proportionally to region size
  - Density emerges in the limit

- Contrast:

  Dirichlet process:
  $$G = \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k} \quad \Rightarrow \quad \text{discrete}$$
  Polya tree (with growing α): continuous.

6.9 Effect of Parameters

- c controls variability around base measure G_0:
  - $c \uparrow \Rightarrow$ tighter around G_0
  - $c \downarrow \Rightarrow$ rougher, more variable draws

6.10 Prior Centering

A Polya tree prior is:
$$G \sim PT(\Pi, \mathcal{A})$$

- Π: recursive partition
- $\mathcal{A} = \{(\alpha_{\epsilon 0}, \alpha_{\epsilon 1})\}$: Beta parameters

Goal:
$$\mathbb{E}[G] = G_0$$

6.11 Centering Method 1: Quantile Partition

- Use symmetric splits:
  $$\alpha_{\epsilon 0} = \alpha_{\epsilon 1}$$
- Define Π using quantiles of G_0:
  - Split at median: $G_0^{-1}(1/2)$
  - Then quartiles: $G_0^{-1}(1/4), G_0^{-1}(3/4)$
  - Continue recursively
- Ensures structure aligns with G_0

6.12 Centering Method 2: Match Split Probabilities

- Fix partition Π
- Choose parameters such that:

$$\frac{\alpha_{\epsilon 0}}{\alpha_{\epsilon 0}+\alpha_{\epsilon 1}} = G_0(B_{\epsilon 0} \mid B_\epsilon)$$

- Equivalent:

$$\alpha_{\epsilon 0} \propto G_0(B_{\epsilon 0} \mid B_\epsilon), \quad \alpha_{\epsilon 1} \propto G_0(B_{\epsilon 1} \mid B_\epsilon)$$

6.13 Why This Centers the Prior

- By definition:

$$Y_\epsilon = \frac{G(B_{\epsilon 0})}{G(B_\epsilon)}$$

- Then:

$$G(B_{\epsilon 0}) = G(B_\epsilon) Y_\epsilon$$

- Taking expectation:

$$\mathbb{E}[G(B_{\epsilon 0}) \mid G(B_\epsilon)] = G(B_\epsilon)\mathbb{E}[Y_\epsilon]$$

- Since:

$$\mathbb{E}[Y_\epsilon] = \frac{\alpha_{\epsilon 0}}{\alpha_{\epsilon 0}+\alpha_{\epsilon 1}} = G_0(B_{\epsilon 0} \mid B_\epsilon)$$

- We get:

$$\mathbb{E}[G(B_{\epsilon 0}) \mid G(B_\epsilon)] = G(B_\epsilon) G_0(B_{\epsilon 0} \mid B_\epsilon)$$

- Recursively:

$$\mathbb{E}[G(B_{\epsilon_1 \cdots \epsilon_m})] = G_0(B_{\epsilon_1 \cdots \epsilon_m})$$

6.14 Posterior Inference

- Data:

$$x_1, \ldots, x_n \mid G \overset{iid}{\sim} G$$

- At node ϵ:

$$Y_\epsilon \sim \text{Beta}(\alpha_{\epsilon 0}, \alpha_{\epsilon 1})$$

- Count data:

$$n_{\epsilon 0} = \#\{x_i \in B_{\epsilon 0}\}, \quad n_{\epsilon 1} = \#\{x_i \in B_{\epsilon 1}\}$$

- Posterior:

$$Y_\epsilon \mid x_{1:n} \sim \text{Beta}(\alpha_{\epsilon 0} + n_{\epsilon 0}, \alpha_{\epsilon 1} + n_{\epsilon 1})$$

6.15 Posterior Predictive Sampling

- To sample x_{n+1}:
- Step 1: traverse tree
  - At node ϵ, sample:
$$Y_\epsilon \mid x_{1:n} \sim \text{Beta}(\alpha^*_{\epsilon 0}, \alpha^*_{\epsilon 1})$$
  - Choose left/right branch
- Step 2: stop at a leaf and sample:
$$x_{n+1} \sim G_0(\cdot \mid B_\epsilon)$$

6.16 Marginal Distribution

- Goal:
$$p(x_1, \ldots, x_n) = \int \prod_{i=1}^{n} G(x_i)\, dP(G)$$
- Key structure:
  - Each x_i follows a path down the tree
  - Contribution depends on shared path with previous points
- Leads to tractable likelihood via Beta-Binomial recursion

6.17 Polya Tree: Marginal Distribution

- The marginal density after integrating out G:
$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} G_0(x_i) \prod_{j=2}^{n} \prod_{m=1}^{m^*(x_j)} \frac{\alpha^*_{\epsilon_m(x_j)}}{\alpha_{\epsilon_m(x_j)}} \cdot \frac{\alpha_{\epsilon_m 0(x_j)} + \alpha_{\epsilon_m 1(x_j)}}{\alpha^*_{\epsilon_m 0(x_j)} + \alpha^*_{\epsilon_m 1(x_j)}}$$
- Notation:
  - $\epsilon_m(x_j)$: index of level-m cell containing x_j
  - $m^*(x_j)$: first level where x_j is isolated
  - α^*: posterior-updated parameters
- Interpretation:
  - Base contribution: $\prod_i G_0(x_i)$
  - Correction: depends on shared paths in the tree

6.18 Mixture of Polya Trees (MPT)

- Issue: single PT produces rough densities (non-differentiable)
- Fix: introduce mixture over centering measures:

$$G \mid \theta \sim PT(\mathcal{A}, G_{0,\theta}), \quad \theta \sim p(\theta)$$

- Effect:
  - Smooths discontinuities
  - Produces continuous predictive densities
- Key idea: averaging over θ removes boundary artifacts

6.19 Motivation: Joint Modeling

- Goal:

$$[T, Y] = (\text{time-to-event, longitudinal response})$$

- Challenge:
  - heterogeneous populations
  - multiple treatments
  - parametric assumptions too restrictive
- Traditional:

$$[T, Y] = [Y][T \mid Y]$$

- Alternative:

$$[T, Y] = [T][Y \mid T]$$

  - $[T]$: nonparametric (Polya tree)
  - $[Y \mid T]$: regression model

6.20 Multivariate Polya Tree

- Centering:

$$G_0 = \mathcal{N}_p(\mu, \Sigma), \quad \Sigma = UU^\top$$

- Construction:
  - Step 1: dyadic partition in $\mathbb{R}$

$$B_0(m, k) = (\Phi^{-1}((k-1)/2^m), \Phi^{-1}(k/2^m)]$$

  - Step 2: form rectangles in $\mathbb{R}^p$

$$B_0(m, \mathbf{k}) = \prod_{j=1}^{p} B_0(m, k_j)$$

  - Step 3: affine transform

$$B(m, \mathbf{k}) = \{\mu + Uz : z \in B_0(m, \mathbf{k})\}$$

- Result:
  - recursive partition aligned with Gaussian geometry
  - partitions $\mathbb{R}^p$ into 2^{mp} cells at level m

6.21 Rubbery Polya Tree

- Problem: standard PT has independent splits $\Rightarrow$ discontinuities
- Idea: introduce dependence across splits
- Construction (finite example):

$$Y_{00} \sim \text{Beta}(\alpha_2, \alpha_2), \quad Z_{00} \mid Y_{00} \sim \text{Binomial}(\delta_{00}, Y_{00})$$

$$Y_{10} \mid Z_{00} \sim \text{Beta}(\alpha_2 + Z_{00}, \alpha_2 + \delta_{00} - Z_{00})$$

- Effect:
  - neighboring splits positively correlated
  - smoother densities

6.22 Linear Dependent Tailfree Process (LDTP)

- Goal: regression version of PT
- Replace fixed splits with covariate-dependent splits:

$$G_x(B_{e0} \mid B_e) = Y_{x,e0}$$

- Model:

$$Y_{x,e0} = \frac{\exp(x^\top \beta_{e0})}{1 + \exp(x^\top \beta_{e0})}$$

- Result:
  - distribution varies smoothly with x
  - predictor affects entire shape of density

6.23 Why LDTP Helps

- Standard regression:
$$Y_i \mid x_i \sim \mathcal{N}(\mu(x_i), \sigma^2(x_i))$$
- Limitation:
  - only mean/variance change
  - shape fixed (Gaussian)
- LDTP:
$$Y_i \mid x_i \sim G_{x_i}$$
- Advantage:
  - full distribution adapts with x
  - captures skewness, multimodality, tail behavior

6.24 LDTP Case Study: Growth Curve Analysis

- Data:
$$Y_i = \text{immunoglobulin concentration}, \quad x_i = \text{age}$$
- Objective:
$$Y_i \mid x_i$$
- Issue:
  - distribution shape evolves with x
  - skewness, tails, modality change
- Model:
$$Y_i = m(x_i) + \epsilon_i, \quad \epsilon_i \sim G_{x_i}$$
- G_x: LDTP prior
- Advantage:
  - models full conditional distribution
  - allows shape to vary with x

6.25 Dependent Multivariate Polya Trees

- Goal:
$$\mathcal{G} = \{G_x : x \in \mathcal{X}\}$$
- Requirement:

  - each G_x is marginally a PT
  - dependence across x
- Replace scalar split:
$$Y_\epsilon \to \{Y_{\epsilon,x} : x \in \mathcal{X}\}$$
- Constraint:
$$Y_{\epsilon,x} \sim \text{Beta}(\alpha_\epsilon, \alpha_\epsilon)$$
- Construction:
$$\{Y_{\epsilon,x}\}_{x\in\mathcal{X}} \sim \text{MBP}(\alpha_0, \alpha_1, Q)$$
- Q: controls dependence across covariates

6.26 Multivariate Beta Process (MBP)

- Goal:
$$\{Y_x : x \in \mathcal{X}\}$$
- Properties:
  - $Y_x \sim \text{Beta}(\alpha_0, \alpha_1)$
  - dependence across nearby x
- Key idea:
  - share Gamma components
- Representation:
$$Y = \frac{G^0}{G^0 + G^1}, \quad G^0, G^1 \sim \Gamma$$
- Dependence:
  - overlapping Gamma terms $\Rightarrow$ correlation

6.27 MBP Construction

- Gamma process on augmented space
- Kernel:
$$Q = \{q_x : x \in \mathcal{X}\}$$
- Regions:
$$S_x^0 = \{(t, y) : 0 < y < \alpha_0 q_x(t)\}, \quad S_x^1 = \{(t, y) : -\alpha_1 q_x(t) < y < 0\}$$

- Define:

$$Y_x = \frac{G(S_x^0)}{G(S_x^0) + G(S_x^1)}$$

- Properties:
  - marginal Beta
  - dependence via overlap of S_x^0, S_x^1

6.28 Finite-Dimensional Representation for MCMC

- Introduce:

$$Z_i \mid Y_{x_i} \sim \text{Bernoulli}(Y_{x_i})$$

- Representation:

$$(\xi_{ij}, \zeta_{ij}) \sim \text{DP}(\nu_{x_1,\dots,x_m})$$

- Sampling:
  - search first atom in region S_{x_i}
  - set $Z_i = \mathbb{I}(\zeta_{ij} > 0)$
- Key result:
  - MBP reduces to DP-based Polya urn
  - tractable MCMC

6.29 Mixture of Polya Trees Revisited

- Model:

$$G \mid \theta \sim PT(\mathcal{A}, G_{0,\theta}), \quad \theta \sim p(\theta)$$

- Effect:
  - smooth densities
  - removes partition artifacts

6.30 Polya Trees in Joint Modeling

- Goal:

$$[T, Y]$$

- Model:

$$[T, Y] = [T][Y \mid T]$$

- Specification:

- T: PT prior
- $Y \mid T$: regression

- Advantage:
  - avoids restrictive parametric survival models
  - captures heterogeneous populations

6.31 Summary of Extensions

- Standard PT:
  - independent splits
  - rough densities
- MPT:
  - mixture over centering measures
  - smoother densities
- LDTP:
  - predictor-dependent splits
  - full conditional density regression
- Dependent PT:
  - preserves Beta marginals
  - uses MBP for dependence
- MBP:
  - induces dependence via shared Gamma processes
  - enables DP-based computation

6.32 Alternative Construction via MCMC

- Goal: introduce branching indicators at a node

$$Z_i \in \{0, 1\}, \quad i = 1, \dots, m$$

- Interpretation:
  - $Z_i = 1 \Rightarrow$ left child
  - $Z_i = 0 \Rightarrow$ right child
- Given split probability at x_i:

$$Z_i \mid Y_{x_i} \sim \text{Bernoulli}(Y_{x_i})$$

- Instead of sampling Y_{x_i} directly:
  - sample induced branching variables

$$Z = (Z_1, \ldots, Z_m)$$

  - sufficient for tree recursion

6.33 DP Representation

- Define region:

$$S = \bigcup_{i=1}^{m} S_{x_i}, \quad S_{x_i} = S_{x_i}^0 \cup S_{x_i}^1$$

- Draw:

$$D_{x_1,\ldots,x_m} \sim DP(\nu_{x_1,\ldots,x_m})$$

- Generate latent points:

$$(\xi_{ij}, \zeta_{ij}) \in S, \quad (\xi_{ij}, \zeta_{ij}) \mid D \overset{iid}{\sim} D_{x_1,\ldots,x_m}$$

6.34 Polya Urn Sampling

- Let $w_r = (\xi, \zeta)$
- Polya urn update:

$$w_r \mid w_1, \ldots, w_{r-1} \sim \frac{\nu(S)}{\nu(S) + r - 1}\nu + \frac{1}{\nu(S) + r - 1}\sum_{\ell=1}^{r-1} \delta_{w_\ell}$$

- For each observation i:

$$j_i = \inf\{j : (\xi_{ij}, \zeta_{ij}) \in S_{x_i}\}$$

- Define indicator:

$$Z_i = \mathbb{I}(\zeta_{ij_i} > 0)$$

- Interpretation:
  - hit in $S_{x_i}^0 \Rightarrow Z_i = 1$
  - hit in $S_{x_i}^1 \Rightarrow Z_i = 0$

6.35 Take-Home Message

- MBP / LDTP sampling reduces to:
  - DP-based Polya urn
  - sampling latent points instead of split probabilities
- Key advantage:
  - avoids direct sampling of $\{Y_{x_i}\}$
  - enables scalable MCMC

Chapter 7

Bayesian Nonparametric Survival Analysis

7.1 Survival Analysis: Setup

- Time-to-event data: model the random event time T.
- Observations (with censoring):

$$Y_i = \min(T_i, C_i), \quad \delta_i = \mathbb{I}(T_i < C_i)$$

- Likelihood (independent censoring):

$$\prod_{i=1}^{n} f(T_i)^{\delta_i} S(C_i)^{1-\delta_i}$$

- Key quantities:

$$S(t) = \Pr(T > t), \quad \lambda(t) = \frac{f(t)}{S(t)}, \quad \Lambda(t) = \int_0^t \lambda(u)\, du$$

7.2 Semiparametric Survival Models

7.2.1 Proportional Hazards (PH)

$$\lambda_x(t) = \lambda_0(t) \exp(x^\top \beta)$$

- $\lambda_0(t)$: baseline hazard (nonparametric)
- $\exp(x^\top \beta)$: multiplicative covariate effect
- Hazard ratio:

$$\frac{\lambda_{x_1}(t)}{\lambda_{x_2}(t)} = \exp((x_1 - x_2)^\top \beta)$$

7.2.2 Accelerated Failure Time (AFT)

$$\log T = x^\top \beta + \epsilon$$

- Covariates rescale time:
$$S_x(t) = S_0(\exp(-x^\top \beta)t)$$
- BNP extension: place a DPM prior on ϵ

7.2.3 Proportional Odds (PO)

$$\frac{S_x(t)}{1 - S_x(t)} = \frac{S_0(t)}{1 - S_0(t)} \exp(-x^\top \beta)$$

- Allows time-varying hazard ratios

7.3 Bayesian Semiparametric PH Models

- Place BNP prior on baseline hazard:
$$\lambda_0(t) \quad \text{or} \quad \Lambda_0(t)$$
- Common priors:
  - Gamma process (continuous hazard)
  - Beta process (discrete jumps)
  - Piecewise exponential models
- Advantage:
  - full posterior over survival curve

7.4 Neutral-to-the-Right (NTR) Processes

- A random CDF F is NTR if
$$1 - F(t_1),\ \frac{1 - F(t_2)}{1 - F(t_1)}, \ldots$$
are independent.
- Equivalent representation:
$$F(t) = 1 - \exp\{-Y(t)\}$$
where $Y(t)$ has independent increments.
- Interpretation:
$$Y(t) = \Lambda(t)$$
(cumulative hazard)

- Key property:
  - conjugacy under censoring
- Special case:
  - Dirichlet process $\subset$ NTR

7.5 Fully Nonparametric Survival Regression

- Goal: model full conditional family

$$\{S(t \mid x) : x \in \mathcal{X}\}$$

- No restriction to:
  - proportional hazards
  - log-linear time scaling

7.5.1 ANCOVA-DDP

- Model:

$$y_i = \log T_i = x_i^\top \beta + \epsilon_i$$

- Error distribution:

$$\epsilon_i \sim G_{x_i}$$

- $\{G_x\}$ follows a dependent Dirichlet process
- Interpretation:
  - full distribution shape varies with x
  - borrowing strength across nearby covariates

7.5.2 Linear Dependent Tailfree Process (LDTP)

- Replace PT splits with covariate-dependent ones:

$$G_x(B_{e0} \mid B_e) = \frac{\exp(x^\top \beta_{e0})}{1 + \exp(x^\top \beta_{e0})}$$

- Produces:

$$G = \{G_x : x \in \mathcal{X}\}$$

- Key property:
  - full conditional density changes shape with x

7.6 Why BNP Survival Models

- Parametric models:
  - too rigid
- Semiparametric models:
  - restrict covariate effects to β
- BNP models:
  - allow
    - multimodality
    - skewness
    - time-varying effects

7.7 Fully Nonparametric Survival Regression: LDTP (Interpretation)

7.7.1 Interpretation I

- **Why fully nonparametric:**
  - Separate parameter β_{e0} at each node e of the infinite tree
  - Covariate effect is distributed across infinitely many parameters
  - The family $\{G_x\}$ can represent arbitrarily complex conditional distributions
- **How covariates enter:**

$$G_x(B_{e0} \mid B_e) = \frac{\exp(x^\top \beta_{e0})}{1 + \exp(x^\top \beta_{e0})}$$

  - Covariates act locally at each resolution level
  - Can reshape distribution differently across the support
  - Allows asymmetric effects (e.g., left vs right tail)

7.7.2 Interpretation II

- **Prior on coefficients:**

$$\beta_{e0} \sim \mathcal{N}\left(0, g(X^\top X)^{-1}\right), \quad g = \frac{2n}{c}$$

  - Shrinks toward zero $\Rightarrow$ equal splits (regularization)
  - Prevents overfitting at deep tree levels
- **Comparison with ANCOVA-DDP:**

- Both define $\{G_x\}$ fully nonparametrically
- ANCOVA-DDP:
  * mixture of Gaussians
  * smooth density
- LDTP:
  * tree-based partition
  * exact (non-mixture) representation
- Different computation and theoretical properties

7.8 Alternative BNP Construction: DDP-GP

7.8.1 Model Construction

- Replace linear mean with a Gaussian process:

$$\theta_h(Z) \sim \mathcal{GP}(\mu_h, C)$$

$$\mu_h(Z_i; \beta_h) = Z_i^\top \beta_h$$

- Covariance kernel:

$$C(Z_i, Z_\ell) = \sigma_0^2 \exp\left(-\sum_{d=1}^{D} \frac{(Z_{id} - Z_{\ell d})^2}{\lambda_d^2}\right) + \delta_{i\ell} J^2$$

7.8.2 Interpretation

- Similar covariates $\Rightarrow$ similar function values
- λ_d: length-scale controls smoothness
- J^2: jitter for numerical stability

7.9 DDP-GP Model for Survival Data

- Data:
  - $Y_i = \log(T_i)$
  - Z_i: covariates
  - δ_i: censoring indicator
- Model:

$$p(y_i \mid Z_i, F) = F_{Z_i}(y_i)$$

- Prior:

$$\{F_Z\} \sim \text{DDP-GP}$$

- Likelihood (right censoring):

$$\mathcal{L}(\theta \mid D_n) = \prod_{i=1}^{n} \left\{ f_{Z_i}(Y_i \mid \theta)^{\delta_i} \left[1 - F_{Z_i}(Y_i \mid \theta)\right]^{1-\delta_i} \right\}$$

- Interpretation:
  - Observed events $\Rightarrow$ density contribution
  - Censored observations $\Rightarrow$ survival contribution

7.10 Comparison of BNP Survival Models

- **ANCOVA-DDP**
  - Mixture-based
  - Smooth densities
  - Clustering interpretation
- **LDTP**
  - Tree-based
  - Explicit control of distribution shape
  - Highly flexible local structure
- **DDP-GP**
  - Function-based dependence
  - Smooth variation over covariates
  - Strong geometric interpretation via kernels